PMATH 352 Complex Analysis

We will start with 2-3 weeks of $\mathbb{R}^2$ discussions, focusing on Green's Theorem and harmonic functions. This will motivate our discussion of complex analytic functions. This will closely follow Flannigan's Complex Variables.

Definition
For $(x, y) \in \mathbb{R}^2$, define $|(x, y)| = \sqrt{x^2 + y^2}$, and for $(u, v) \in \mathbb{R}^2$ as well, define the distance from (x, y) to (u, v) to be $|(x, y) - (u, v)|$.

Definition
Let $z_o \in \mathbb{R}^2$ and let $r > 0 \in \mathbb{R}$. The open disc of radius r centered at z_o is

$$D(z_o; r) = \{z \in \mathbb{R}^2 : |z - z_o| < r\}$$

and the closed disc of radius r centered at z_o is

$$\overline{D}(z_o; r) = \{z \in \mathbb{R}^2 : |z - z_o| \leq r\}$$

Definition
Let $\Omega \in \mathbb{R}^2$ and let $z \in \mathbb{R}^2$.
Definition
Let $\Gamma \subseteq \mathbb{R}^2$. We say that Γ is a curve if and only if there exists a closed interval $[a, b] \subseteq \mathbb{R}$ and a function $\alpha : [a, b] \to \Gamma$ mapping $t \mapsto (\alpha_1(t), \alpha_2(t))$ such that (i) α is continuous and (ii) α is surjective. We refer to α as the parameterization of Γ.

Definition
A curve (Γ, α) is called smooth if and only if:
(iii) $\alpha_1'(t)$ and $\alpha_2'(t)$ exist and are continuous for all $t \in (a, b)$ and one-sided differentiable at the end-points.
(iv) for every $t \in [a, b]$, $\alpha'(t) \neq (0, 0)$
(v) if $\alpha(a) = \alpha(b)$ (i.e. Γ is closed), then $\alpha'(a) = \alpha'(b)$.

Definition
We say that (Γ, α) is piecewise smooth if there exist finitely many $a = a_0 < \cdots < a_n = b$ such that $\Gamma_k = \{\alpha(t) : t \in [a_k, a_{k+1}]\}$ is smooth for each $k = 0, \ldots, n-1$.

Definition
For a smooth curve (Γ, α), we refer to the vector $T(s) = \alpha'(s) = (\alpha_1'(s), \alpha_2'(s))$ as the tangent vector or velocity vector.

Definition
We say that (Γ, α) is simple if and only if $\alpha |_{(a,b)}$ is one-to-one. In other words, Γ does not self-intersect.

Definition
Let $\alpha : [a, b] \to \Gamma$ be a piecewise-smooth parameterization of the curve Γ. The length of (Γ, α). The length of (Γ, α) is given by length$(\alpha) = \int_a^b |\alpha'(t)| dt$.

Theorem (Parameterization by arc length)
Let Γ be a piecewise-smooth curve of length $L > 0$. There exists a piecewise-smooth parameterization $\sigma : [0, L] \to \Gamma$ such that for every $s \in [0, L]$:
(i) the tangent vector $\sigma'(s)$ has unit length, $|\sigma'(s)| = 1$
(ii) the distance traveled along Γ from $\sigma(0)$ to $\sigma(s)$ is exactly s.

Example:
A circle C_r of radius $r > 0$ can be parameterized by arc length via the map $\sigma : [0, 2\pi r] \to C_r$ given by $\sigma(s) = (r\cos(\frac{s}{r}), r\sin(\frac{s}{r}))$.

An algorithm for finding this is as follows: take any old parameterization $\alpha : [a, b] \to \Gamma$. Define $F(\tau) = \int_a^\tau |\alpha'(t)|dt$. This gives $F : [a, b] \to [0, L]$. F is monotonically increasing on $[a, b]$, so it is invertible. Now just invert it to get $F^{-1} : [0, L] \to [a, b]$, so take $\alpha \circ F^{-1}$. (Formalizing this gives a proof)

Definition
An open subset Ω of $\mathbb{R}^2$ is disconnected if and only if there exists non-empty open subsets Ω_1, Ω_2 of $\mathbb{R}^2$ such that $\Omega_1 \cup \Omega_2 = \Omega$ and $\Omega_1 \cap \Omega_2 = \emptyset$. Otherwise, it is called connected.

Definition
An open and connected subset of $\mathbb{R}^2$ is called a domain.

Definition
A piecewise-smooth, simple, closed curve is called a Jordan curve.

Theorem (Jordan)
Let Γ be a Jordan curve. Then $\mathbb{R}^2 \setminus \Gamma$ consists of two disjoint domains:
– One bounded ("inside"), and
– One unbounded ("outside").
Both domains have Γ as their boundary. Furthermore, if a point on the inside is joined by a path to a point on the outside, then the path crosses Γ.

Definition
A (k-connected) Jordan domain is a bounded domain Ω whose boundary $\partial\Omega$ is the union of $k \geq 1$ distinct Jordan curves, each parameterized so that, as a point moves in the direction of parameterization, the domain Ω is always lying to its left (orientation).

Definition
Let Γ be a Jordan domain, and let Γ be one of the Jordan curves in $\partial\Omega$. Let $\sigma : [0, L] \to \Gamma$ be a parameterization of Γ by arc length. Then $T(s) = (\sigma_1'(s), \sigma_2'(s))$ has unit length. We define the outward normal vector to be $N(s) = (\sigma_2'(s), -\sigma_1'(s))$. This has three properties:
(i): $|N(s)| = 1$
(ii): $N(s)$ is orthogonal to $T(s)$
(iii): $N(s)$ points outward, away from Ω.

Definition
Let Ω be a domain, and $f : \Omega \to \mathbb{R}$. $f \in C^0(\Omega)$ means it is continuous, $f \in C^k(\Omega)$ means it is k-times continuously partially differentiable.

Definition
Let $f \in C^1(\Omega)$. Then $\nabla f = (\frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f)$. Geometrically it represents the direction and rate of steepest ascent.

Definition
Let v be a unit vector, $f \in C^1(\Omega)$, $z \in z$. The directional derivative of f at z in the direction of v is $f'(z; v) = \frac{d}{ds} f(z + sv)|_{s=0} = \nabla f(z) \cdot v$.

Exercise
Prove that $-|\nabla f(z)| \leq f'(z; v) \leq |\nabla f(z)|$.

Note:
We will be interested in directional derivatives in the direction of an outward normal vector $N(s)$.

Some Integral Calculus:
Let Γ be a piecewise-smooth curve in $\mathbb{R}^2$. Let $p(x,y)$ and $q(x,y)$ be continuous in some set containing Γ. We would like to compute the line integral

$$I = \int_\Gamma (pdx + qdy)$$

We can do this by choosing a parametrization $\alpha : [a,b] \to \Gamma$, $\alpha(t) = (\alpha_1(t), \alpha_2(t))$. Then $x = \alpha_1(t)$ and $y = \alpha_2(t)$, so $\frac{dx}{dt} = \alpha_1'(t)$ and $\frac{dy}{dt} = \alpha_2'(t)$. We define

$$I = I_\alpha = \int_a^b (p(\alpha(t))\alpha_1'(t) + q(\alpha(t))\alpha_2'(t))dt$$

Definition
We say that two paramterizations $\alpha : [a_1, b_1] \to \Gamma$ and $\beta : [a_2, b_2] \to \Gamma$ are equivalent if and only if there exists a one-to-one and onto continuous piecewise differentiable function $E : [a_1, b_1] \to [a_2, b_2]$ such that $\alpha(t) = \beta(E(t))$ for all $t \in [a_1, b_1]$ and $E'(t) > 0$ for all $t \in [a_1, b_1]$.

Exercise
Prove that if α and β are two equivalent parametrizations, then $I_\alpha = I_\beta$.
Proof:
We have

$$I_\alpha = \int_{a_1}^{b_1} p(\alpha(t))\alpha'(t)dt = \int_{a_1}^{b_1} p(\beta(E(t)))\beta'(E(t))E'(t)dt = \int_{a_2}^{b_2} p(\beta(u))\beta'(u)du$$

Exercise
Compute each of the following over the unit circle C_1 (counterclockwise orientation):
(a) $\int_{C_1} dy$
(b) $\int_{C_1} xdx - ydy$
(c) $\int_{C_1} x(dx + dy)$
(d) $\int_{C_1}$????

Definition
Let Ω be a domain and let $u \in C^1(\Omega)$. The differential of u is

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

What happens if we say $p = \frac{du}{dx}$, $q = \frac{du}{dy}$ in our line integral?

Theorem (Line Integral of Exact Differentials)
Let Γ be a piecewise-smooth curve from z_0 to z_1 lying inside a domain Ω. Let $u \in C^1(\Omega)$. Then

$$\int_\Gamma du = u(z_1) - u(z_0)$$

Therefore, when Γ is closed this integral is 0.
Proof:
We have $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$. So

$$\int_\Gamma du = \int_\Gamma (\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy)$$

Parametrize this with $\alpha(t) = (\alpha_1(t), \alpha_2(t))$, giving

$$= \int_a^b (\frac{\partial}{\partial x} u(\alpha_1(t), \alpha_2(t))\alpha_1'(t) + \frac{\partial}{\partial y} u(\alpha_1(t), \alpha_2(t))\alpha_2'(t))dt$$

$$= \int_a^b (\frac{d}{dt} u(\alpha_1(t), \alpha_2(t)))dt$$

$$= u(\alpha_1(t), \alpha_2(t)) \mid_a^b = u(z_1) - u(z_o) \quad \blacksquare$$

Exercise
If $du = pdx + qdy$ for $u \in \mathcal{C}^2(\Omega)$, then $p_y = q_x$, so $u_x = p$ and $u_y = q$, so by Clairaut's Theorem $p_y = q_x$.

Theorem (Green):
Let Ω be a k-connected Jordan domain, and suppose that $p, q \in C^1(\Omega^+)$, where Ω^+ is a domain such that $\Omega \cup \partial\Omega \subseteq \Omega^+$ and $\partial\Omega$ is positively oriented. Then

$$\int_{\partial\Omega} (pdx + qdy) = \iint_\Omega (q_x - p_y)dxdy$$

Proof (rectangle case):
DIAGRAM OF RECTANGLE. ITS BOTTOM LINE IS Γ_1, ITS TOP LINE IS Γ_1. Left/right lines are Γ_2 and Γ_4.

It is sufficient to show that

$$\int_{\partial\Omega} pdx = -\iint_\Omega p_y dxdy$$

But

$$-\iint_\Omega p_y dxdy = \int_{x_0}^{x_1}\int_{y_0}^{y_1} p_y(x,y)dydx = -\int_{x_0}^{x_1} p(x,y) \mid_{y_0}^{y_1} dx = -\int_{x_0}^{x_1} (p(x,y_1) - p(x,y_0))dx$$

$$= \int_{x_0}^{x_1} p(x,y_0)dx + \int_{x_1}^{x_0} p(x,y_1)dx$$

$$= \int_{\Gamma_1} pdx + \int_{\Gamma_3} pdx$$

But $\int_{\Gamma_2} pdx = \int_{\Gamma_4} pdx = 0$ because x doesn't change on a line going straight up/down! So this equals

$$= \int_{\partial\Omega} pdx \quad \blacksquare$$

Notation
Let $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ denote the Laplace operator. We are interested in the integral

$$\int_{\partial\Omega} \underbrace{\frac{\partial q}{\partial n}}_{\text{outward normal derivative at } (\sigma_1(s), \sigma(2))} ds$$

By definition,

$$\frac{\partial q}{\partial n} ds = \nabla q \cdot N(s)ds$$

$$= \begin{bmatrix} q_x \\ q_y \end{bmatrix} \cdot \begin{bmatrix} \sigma_2'(s) \\ -\sigma_1'(s) \end{bmatrix} ds = q_x \sigma_2'(s)ds - q_y \sigma_1'(s)ds = q_x dy - q_y dx$$

So,

$$\int_{\partial\Omega} \frac{\partial q}{\partial n} ds = \int_{\partial\Omega} q_x dy - q_y dx$$

Identity #1
Let Ω be a k-connected Jordan domain and let $p, q \in C^2(\Omega+)$. Then

$$\iint_\Omega \nabla p \cdot \nabla q dxdy = \int_{\partial\Omega} p\frac{\partial q}{\partial n} ds - \iint_\Omega p\Delta q dxdy$$

This is a version of integration by parts in $\mathbb{R}^2$!
Proof:

$$\int_{\partial\Omega} p\frac{\partial q}{\partial n} ds = \int_{\partial\Omega} [pq_x dy - pq_y dx] = \int_{\partial\Omega} (-pq_y)dx + (pq_x)dy$$

$$\overbrace{=}^{GT} \iint_\Omega \left(\frac{\partial}{\partial x}(pq_x) + \frac{\partial}{\partial y}(pq_y)\right) dxdy = \iint_\Omega (p_x q_x + pq_{xx} + p_y q_y + pq_{yy})dxdy$$

$$= \iint_\Omega (p_x q_x + p_y q_y + p(q_{xx} + q_{yy}))dxdy = \iint_\Omega (\nabla p \cdot \nabla q + p\Delta q)dxdy \quad \blacksquare$$

Identity #2
Let Ω be a k-connected Jordan domain and let $p, q \in C^2(\Omega^+)$. Then

$$\int_{\partial\Omega} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds = \iint_\Omega (p\Delta q - q\Delta p)dxdy$$

Proof:
Quiz 4 (write down Identity 1 for p, q then write it down for q, p). $\blacksquare$

Inside-Outside Theorem:
Let Ω be a k-connected Jordan domain and let $q \in C^2(\Omega^+)$. Then

$$\int_{\partial\Omega} \frac{\partial q}{\partial n} ds = \iint_\Omega \Delta q dxdy$$

Proof:
Use Identity 1 with p constant. $\blacksquare$

Identity #3
Let Ω be k-connected Jordan domain $\zeta \in \Omega$ and let $q \in C^2(\Omega^+)$. Write $z = (x, y)$ and $r = |z - \zeta|$. Then

$$q(\zeta) = \frac{1}{2\pi}\iint_\Omega \log(r)\Delta q(z)dxdy - \frac{1}{2\pi}\int_{\partial\Omega}\left(\log(r)\frac{\partial q}{\partial n}(z) - q(z)\frac{\partial \log(r)}{\partial n}\right) ds$$

When the Laplacian vanishes, this means the function is uniquely determined by its behavior on the boundary!!!!!!!
Proof:
Below

Definition:
We really care about the Laplacian vanishing! So: Let Ω be a domain in $\mathbb{R}^2$. We say that $u \in C^2(\Omega)$ is harmonic if and only if $\Delta u(x, y) = 0$ for all $(x, y) \in \Omega$.

Why should we care? Plug $\Delta u = 0$ into above identities, but also:

Let $f : \Omega \to \mathbb{C}$ be a complex differentiable function:

$$f'(z_o) := \lim_{z\to 0} \frac{f(z_o + z) - f(z_o)}{z}$$

This limit exits for all $z_o \in \Omega$. This z can approach 0 via really bizarre paths!
We will show that $f(x+iy) = u(x,y) + iv(x,y)$ where both $u(x,y)$ and $v(x,y)$ are harmonic.

To prove identity 3, we will need a lemma.

Lemma:
Fix $\zeta \in \mathbb{R}^2$. Let $r(z) = |z-\zeta|$. Then $\log(r)$ is constant on $C(\zeta; r)$ (the circle of radius r centered at ζ), and $\log(r)$ is harmonic $\mathbb{R}^2 \setminus \{\zeta\}$. That is, $\Delta \log r(z) = 0$ for all $z \in \mathbb{R}^2 \setminus \{\zeta\}$.
Proof (of Lemma and Identity 3):
The first statement is an exercise, the latter statement is on the homework. Otherwise, let $\varepsilon > 0$ and put $\Omega_\varepsilon = \Omega \setminus \overline{D}(\zeta;\varepsilon)$. Then $\partial\Omega_\varepsilon = \partial\Omega \cup C(\zeta;\varepsilon)$. Now write down Identity 2 for Ω_ε:

$$\int_{\partial\Omega_\varepsilon} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds = \iint_{\Omega_\varepsilon} (p\Delta q - q\Delta p)dxdy$$

Set $p = \log r$, where $r = |z-\zeta|$. By the lemma, p is harmonic on Ω_ε, hence $\Delta p = 0$ and we have

$$\int_{\partial\Omega_\varepsilon} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds = \iint_{\Omega_\varepsilon} p\Delta q dxdy$$

Now since $\partial\Omega_\varepsilon = \partial\Omega \cup C(\zeta;\varepsilon)$, we have

$$\int_{\partial\Omega_\varepsilon} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds = \int_{\partial\Omega} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds + \int_{C(\zeta;\varepsilon)} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds$$

Looking only at the right term,

$$\int_{C(\zeta;\varepsilon)} (p\frac{\partial q}{\partial n} - q\frac{\partial p}{\partial n})ds = \int_{C(\zeta;\varepsilon)} (\log r\frac{\partial q}{\partial n} - q\frac{\partial \log r}{\partial n})ds$$

$$= \int_{C(\zeta;\varepsilon)} (\log\varepsilon\frac{\partial q}{\partial n} + q\frac{\partial \log r}{\partial r})ds = \int_{C(\zeta;\varepsilon)} (\log\varepsilon\frac{\partial q}{\partial n} + \frac{q}{r})ds = \int_{C(\zeta;\varepsilon)} (\log\varepsilon\frac{\partial q}{\partial n} + \frac{q}{\varepsilon})ds$$

Now take the arc length parametrization $\sigma : [0, 2\pi\varepsilon] \to C(\zeta;\varepsilon)$ given by $\sigma(s) = (\varepsilon\cos\frac{s}{\varepsilon}, \varepsilon\sin\frac{s}{\varepsilon}) + \zeta$ (centering at ζ). Thus our above integral is

$$\int_0^{2\pi\varepsilon} (\log\varepsilon\frac{\partial q(\sigma_1(s),\sigma_2(s))}{\partial n} + \frac{q(\sigma_1(s),\sigma_2(s))}{\varepsilon})ds = \int_0^{2\pi} (\log\varepsilon\frac{\partial q}{\partial n} + \frac{q}{\varepsilon})\varepsilon d\theta$$

$$= \varepsilon\log\varepsilon\int_0^{2\pi} \frac{\partial q}{\partial n}d\theta + \int_0^{2\pi} q(\varepsilon,\theta)d\theta$$

Then by the MVT,

$$= \underbrace{\varepsilon\log\varepsilon}_{\to 0}\underbrace{\int_0^{2\pi} \frac{\partial q}{\partial n}d\theta}_{\text{constant}} + \underbrace{q(\varepsilon,\theta_\varepsilon)}_{\to q(\zeta)}\underbrace{\int_0^{2\pi} d\theta}_{2\pi} \quad \blacksquare$$

Circumferential Mean Value Theorem
Let Ω be a k-connected Jordan domain. Let $u \in C^2(\Omega^+)$ be harmonic. Let $\zeta \in \Omega$, let $R > 0 \in \mathbb{R}$ such that $\overline{D}(\zeta;R) \subseteq \Omega$. Then,

$$u(\zeta) = \frac{1}{2\pi R}\int_{C(\zeta;R)} u ds = \frac{1}{2\pi}\int_0^{2\pi} u(R,\theta)d\theta$$

Proof:
By Green's identity 3,

$$u(\zeta) = -\frac{1}{2\pi}\int_{C(\zeta;R)} \left(\log R\frac{\partial u}{\partial n}(z) - u(z)\frac{\partial \log r}{\partial n}\right) ds$$

$$= -\frac{1}{2\pi}\int_{C(\zeta;R)} \left(\log R\frac{\partial u}{\partial n}(z) - \frac{u(z)}{R}\right) ds$$

$$= -\frac{R}{2\pi} \underbrace{\int_{C(\zeta;R)} \frac{\partial u}{\partial n}(z)ds}_{=0 \text{ by inside outside thm}} + \frac{1}{2\pi R}\int_{C(\zeta;R)} u(z)ds \quad \blacksquare$$

Solid Mean Value Property
Let $R > 0$, u harmonic in an open set Ω, let $\zeta \in \Omega$, and suppose that $\overline{D}(\zeta; R) \subseteq \Omega$ for some $R > 0$. Then u has a solid mean value property:

$$u(\zeta) = \frac{1}{\pi R^2}\iint_{D(\zeta;R)} udxdy$$

Proof:
We know that u has circumferential mean value property: for every $0 < r \leq R$,

$$u(\zeta) = \frac{1}{2\pi r}\int_0^{2\pi r} uds \overbrace{=}^{s=r\theta} \frac{1}{2\pi}\int_0^{2\pi} ud\theta$$

Multiply both sides by rdr and integrate:

$$\int_0^R u(\zeta)rdr = \frac{1}{2\pi}\int_0^R\int_0^{2\pi} u \underbrace{rdrd\theta}_{dxdy \text{ in cyl. coords}}$$

$$\implies u(\zeta)\frac{R^2}{2} = \frac{1}{2\pi}\iint_{D(\zeta;R)} udxdy \quad \blacksquare$$

Characterization of Harmonic Functions
Let Ω be a domain and let $u \in C^2(\Omega)$. Then u is harmonic iff u has the circumferential mean value property if and only if $\int_\Gamma \frac{\partial u}{\partial n} ds = 0$ for every Jordan curve $\Gamma \subseteq \Omega$

Strong Maximum Principle
Let Ω be a domain and let $u : \Omega \to \mathbb{R}$. We say that u assumes its maximum in Ω if there exists $\zeta \in \Omega$ and $c \in \mathbb{R}$ such that $u(\zeta) = c$ and $u(z) \leq c$ for all $z \in \Omega$. An analogous definition can be stated for a minimum. Then, a nonconstant harmonic function $u : \Omega \to \mathbb{R}$ does not assume its maximum on Ω. Similarly for a minimum.
Proof:
Suppose u does not attain its maximum on Ω. Let this maximum be $c \in \mathbb{R}$ and suppose that $u(\zeta) = c$ for some $\zeta \in \Omega$. Then $A = \{z \in \Omega : u(z) < c\} = u^{-1}((-\infty, c))$ is open, since we are considering the preimage where u is continuous. Since Ω is open and connected (therefore path connected), we can connect ζ to some $z_o \in A$ via a piecewise smooth path $\alpha : [0, 1] \to \Gamma$ such that $\alpha(0) = \zeta$ and $\alpha(1) = z_o$. Since $u(\alpha(0)) = c$ and $u(\alpha(1)) < c$, and α is continuous, there $\exists t \in [0, 1)$ such that $u(\alpha(t)) = c$ and $u(\alpha(t + \varepsilon)) < c$ for ε sufficiently small. Put $\zeta_1 = \alpha(t)$. Then we can find a closed disc $\overline{D}(\zeta_1; R)$ for small enough $R > 0$ such that $\overline{D}(\zeta_1; R) \subseteq \Omega$ and $\overline{D}(\zeta_1; R) \cap A \neq \emptyset$. Let $z_1 \in \overline{D}(\zeta_1; R) \cap A$. Then z_1 lies on the circle $C(\zeta_1; R)$ and $u(z_1) < c$. Because u is continuous, $u(z_1) < c$ for all z on some arc of the circle $C(\zeta_1; R)$, which contains z_1. Now,

$$u(\zeta_1) = \frac{1}{2\pi R}\int_{C(\zeta;R)} u(z)ds \underbrace{<}_{u(z)<u(\zeta_1) \text{ on some arc}} \frac{1}{2\pi R}\int_{C(\zeta_1;R)} u(\zeta_1)ds = u(\zeta_1)$$

where the first equality follows by the CMVT, contradiction. $\blacksquare$

Weak Maximum Principle
Let Ω be a bounded domain with u continuous on $\overline{\Omega}$ and harmonic on Ω. Then u is constant on $\overline{\Omega}$ or

attains it maximum and minimum on $\partial\Omega$ only.
Proof:
A1. $\blacksquare$

Theorem (Harnack's Inequality)
Let $z_o \in \mathbb{R}^2$ and $R > 0$. Let u be harmonic on $D = D(z_o; R)$, and assume that $u(z) \geq 0$ for all $z \in D$. Then for all $z \in D$,

$$0 \leq u(z) \leq \left(\frac{R}{R - |z - z_o|}\right)^2 u(z_o)$$

Proof:
By SMVT,

$$u(z) = \frac{1}{\pi(R - |z - z_o|)^2} \iint_{D'} u dx dy \underbrace{\leq}_{\text{non-neg}} \frac{1}{\pi(R - |z - z_o|)^2} \iint_D u dx dy$$
$$= \frac{\pi R^2}{\pi(R - |z - z_o|)^2} \underbrace{\frac{1}{\pi R^2} \iint_D u dx dy}_{=u(z_o) \text{ by SMVT}} = \frac{R^2}{(R - |z - z_o|)^2} u(z_o)$$

where D and D' are discs surrounding z and z_o. $\blacksquare$

Definition
We say that a harmonic function is entire if and only if its domain is $\mathbb{R}^2$.

Theorem (Liouville)
If u is entire harmonic and is bounded either from above or below, then u is constant.
Proof:
Suppose that $u(z)$ is bounded from below, so there exists $c \in \mathbb{R}$ such that $u(z) \geq c$ for all $z \in \mathbb{R}$. Let $v(z) = u(z) - c$. Let $z, \zeta \in \mathbb{R}^2$ be distinct. Let $R \in \mathbb{R}$ such that $R > |z - \zeta|$. By Harnack's inequality,

$$v(z) \leq \frac{R^2}{(R - |z - \zeta|)^2} v(\zeta) \underbrace{\rightarrow}_{R \to \infty} v(\zeta)$$

hence $v(z) \leq v(\zeta)$. Interchange z and ζ in the above proof to get $v(\zeta) \leq v(z)$, so $v(z) = v(\zeta)$ for arbitrary $z, \zeta \in \mathbb{R}^2$. The proof is analogous for bounding it by above (take $-v(z)$). $\blacksquare$

Complex Numbers - Definition
A complex number is an element of the form $z = x + yi$ for $x, y \in \mathbb{R}$ and i is a formal symbol. This is just a different way to represent a point in $\mathbb{R}^2$. We denote the set of complex numbers as $\mathbb{C}$.

Fact
$\mathbb{C}$ is a field under the following operations of $+$ and $\cdot$: for $z = a + bi$ and $w = c + di$,
$z + w := (a + c) + (b + d)i$
$z \cdot w = (ac - bd) + (ad + bc)i$

The latter implies $i^2 = -1$ and commutativity of $\mathbb{C}$'s multiplication.

Definitions
For $z = x + yi$, $x, y \in \mathbb{R}$, define $\text{Re}(z) = x$, $\text{Im}(y) = y$, $\overline{z} = x - yi$, $|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2}$, and $z^{-1} = \frac{\overline{z}}{|z|^2}$, and $\arg z$ to be the unique angle θ, $-\pi < 0 \leq \pi$ such that $x = |z| \cdot \cos\theta$ and $y = |z| \cdot \sin\theta$. Equivalently, we can define $\theta = \text{atan2}(y, x)$.

We can convert $z = x + yi$, $z \neq 0$, to polar coordinates via $x = r\cos\theta$ and $y = r\sin\theta$ where $r > 0$ and $-\pi < \theta \leq \pi$, which is unique under our definition. Notice that $r = |z|$. Hence $x + yi =$

$|z| \cdot \cos\theta + i|z| \cdot \sin\theta = |z| \underbrace{(\cos\theta + i\sin\theta)}_{\text{point on the unit circle}}.$

Theorem
If $z = |z|(\cos\alpha + i\sin\alpha)$, $w = |w|(\cos\beta + i\sin\beta)$, then $z \cdot w = |zw| = (\cos(\alpha+\beta) + i\sin(\alpha+\beta))$.

Theorem (De Moivre)
For every $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Definition
Let Ω be a subset of $\mathbb{C}$. A complex function is a map $f : \Omega \to \mathbb{C}$ that assigns to every element $z \in \Omega$ exactly one complex number $f(z)$.
Examples
Every function $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{C}$ is an example. Such as $f(z) = (1+i) \cdot z$, $g(z) = z^3$, $h(z) = |z|$, $t(z) = \overline{z}$.

Definition
Every complex function f can be written as $f(z) = u(z) + iv(z)$ where $u : \Omega \to \mathbb{R}$ and $v : \Omega \to \mathbb{R}$ are real and imaginary parts of $f : u = \mathrm{Re}(f), v = \mathrm{Im}(f)$.

Definition
Let $f : \Omega \to \mathbb{C}$ be a complex function, $z_o \in \mathbb{C}$, $w_o \in \mathbb{C}$. We say that $\lim_{z \to z_o} f(z) = w_o$ if and only if for all $\varepsilon > 0$, there exists $\delta > 0$, such that $z \neq z_o \in D(z_o; \delta) \cap \Omega \implies f(z) \in D(w_o; \varepsilon)$. Equivalently, this means that for all $z \in \Omega$, $0 < |z - z_o| < \delta \implies |f(z) - w_o| < \varepsilon$.

Definition
We say that the complex function $f : \Omega \to \mathbb{C}$ is continuous at z_o if and only if
(i) $z_o \in \Omega$, so that $f(z_o)$ is defined
(ii) $\lim_{z \to z_o} f(z) = f(z_o)$.
We say that f is continuous on Ω if and only if it is continuous at every point $z_o \in \Omega$.

Some Extra References
We will start to follow Marsden & Hoffman, Basic Complex Analysis, and Lang's Complex Analysis.

Fact
A complex function $f(z) = u(x, y) + iv(x, y)$ is continuous if and only if u and v are.

Definition
Let $f : \Omega \to \mathbb{C}$ be a complex function, $z_o \in \Omega$. We define the first derivative $f'(z_o)$ of f at z_o by

$$f'(z_o) = \lim_{z \to z_o} \frac{f(z) - f(z_o)}{z - z_o}$$

provided this limit exists a finite number, alternatively denoted $\frac{df}{dz}$.

Properties
All identities from ordinary real calculus are in place, the usual rules surrounding polynomials, the product rule, the quotient rule, and chain rule.

Definition
Let $f : \Omega \to \mathbb{C}$ be a complex function. If $f'(z_o)$ exists at all points $z_o \in \Omega$, we refer to the function $f' : \Omega \to \mathbb{C}$, $z \mapsto f'(z)$ as the derivative of f.

Definition

We say that f is analytic (equivalently, holomorphic, regular, continuously complex differentiable) at z_o if and only if
(i) $f'(z)$ exists for all points z in some open set containing z_o (so $f'(z_o)$ exists, and
(ii) $f'(z)$ is a continuous function of z in some open set containing z_o. We say that f is analytic on Ω if and only if it is analytic at every $z_o \in \Omega$.

Theorem (Cauchy-Riemann Equations)

If $f : \Omega \to C$, $f(z) = u(z) + v(z)i$ is analytic at $z \in \Omega$. then

$$f'(z) = u_x(z) + v_x(z)i = v_y(z) - u_y(z)i$$

so that

$$u_x = v_y$$

and

$$u_y = -v_x$$

Conversely, if u and $v \in \mathcal{C}^1(\Omega)$ and satisfy the above two equations for all $z \in \Omega$, then $f = u + vi$ is analytic in Ω.
Proof:
In the forward direction, let $z = x + yi$ approach $z_o = x_o + y_oi$ along the horizontal line $y = y_o$. Then $z = x + y_oi$ and $z - z_o = (x + y_oi) - (x_o + y_o)i = x - x_o$.

$$\lim_{z\to z_o} \frac{f(z) - f(z_o)}{z - z_o} = \lim_{x\to x_o} \frac{u(x, y_o) - u(x_o, y_o)}{x - x_o} + \lim_{x\to x_o} \frac{v(x, y_o) - v(x_o, y_o)}{x - x_o} i = u_x(x_o, y_o) + v_x(x_o, y_o)i$$

Now let $z = x + yi$ approach along the vertical line $x = x_o$. Then $z = x_o + yi$ and $z - z_o = (x_o + yi) - (x_o + y_o)i = i(y - y_o)$.

$$\lim_{z\to z_o} \frac{f(z) - f(z_o)}{z - z_o} = \lim_{y\to y_o} \frac{u(x_o, y) - u(x_o, y_o)}{i(y - y_o)} + \lim_{y\to y_o} \frac{v(x_o, y) - v(x_o, y_o)}{i(y - y_o)} i$$

$$= \lim_{y\to y_o} \frac{v(x_o, y) - v(x_o, y_o)}{y - y_o} - \lim_{y\to y_o} \frac{u(x_o, y) - u(x_o, y_o)}{y - y_o} i = v_y(x_o, y_o) - u_y(x_o, y_o)i$$

Conversely, let $u, v \in \mathcal{C}^1(\Omega)$ satisfy the Cauchy-Riemann equations. By Taylor's Theorem,

$$u(z) = u(z_o) + u_x(z_o)(x - x_o) + u_y(z_o)(y - y_o) + O(|z - z_o|^2)$$

and

$$v(z) = v(z_o) + v_x(z_o)(x - x_o) + v_y(z_o)(y - y_o) + O(|z - z_o|^2)$$

Thus, for $f(z) = u(z) + v(z)i$,

$$f(z) = f(z_o) + (u_x(z_o) + v_x(z_o)i) + (u_y(z_o) + v_y(z_o)i)(y - y_o) + O(|z - z_o|^2)$$

$$\overbrace{=}^{CR} f(z_o) + (u_x + v_xi)(x - x_o) + (-v_x + u_xi)(y - y_o) + O(|z - z_o|^2)$$
$$= f(z_o) + (u_x(z_o) + v_x(z_o)i)(z - z_o) + O(|z - z_o|^2)$$

Hence

$$\frac{f(z) - f(z_o)}{z - z_o} = u_x(z_o) + v_x(z_o)i + \frac{O(|z - z_o|^2}{z - z_o}$$

giving

$$\left| \frac{f(z) - f(z_o)}{z - z_o} - (u_x(z_o) + v_x(z_o)i) \right| = O(|z - z_o|) \overset{z\to z_o}{\to} 0$$

Hence
$$\lim_{z\to z_o} \frac{f(z)-f(z_o)}{z-z_o} = u_x(z_o) + v_x(z_o)i$$
where continuity holds because u_x and v_x are continuous. ■

There are also polar coordinates versions of the Cauchy-Riemann equations (provided the argument is defined), $u_r = \frac{1}{r}v_\theta$, $v_r = -\frac{1}{r}u_\theta$.

Example
Let $f(z), g(z) \neq 0$ be complex polynomials. Define $\Omega = \mathbb{C} \setminus \{z : g(z) = 0\}$. Then $\frac{f(z)}{g(z)}$ is analytic on Ω. The proof for this is analogous to its real equivalent.

Definition
We define the complex exponential $\exp : \mathbb{C} \to \mathbb{C}$ by $\exp(x+yi) := e^x(\cos y + i \sin y)$. Hence $u(x,y) = e^x \cos y$, $v(x,y) = e^x \sin(y)$. By the quiz of lecture 10, these real and imaginary parts satisfy the Cauchy-Riemann equations, and hence exp is analytic.
Also, from the theoreom on Cauchy-Riemann equations we know that $\frac{d}{dz}\exp(z) = u_x(z) = v_x(z)i = e^x\cos(y) + e^x\sin(y)i = \exp(z)$.
When $z = \pi i$, we obtain Euler's formula $e^{\pi i} = -1$.
As some examples, we get $\exp(i) = \cos(1) + i\sin(1)$, $\exp(i\pi/2) = i$, $\exp(\log 2 - i\pi/2) = -2i$.

Remark
This definition is very natural: when defining exp, we want the identity $\exp(x+yi) = \exp(x)\exp(yi)$ to hold. But write $e^{yi} = u(y) + v(y)i$, and write $\frac{d^2}{dy^2}e^{yi} = \frac{d}{dy}(ie^{yi}) = -e^{yi}$, hence $u''(y) = -u(y)$, $v''(y) = -v(y)$, which tells us that $u(y) = A\cos(y) + B\sin(y)$ and $v(y) = C\cos(y) + D\sin(y)$. But with the boundary conditions $e^{0i} = 1$, $u(0) = 1$, $v(0) = 0$, we get that $A = 1, B = 0, C = 0, D = 1$.

Properties of the Complex Exponential
(i) $\forall z_1, z_2 \in \mathbb{C}$, $\exp(z_1 + z_2) = \exp(z_1)\exp(z_2)$.
(ii) $\mathrm{im}(\exp(z)) = \mathbb{C} \setminus \{0\}$, since e^x scales a point $(\cos y + i \sin y)$ on the unit circle by an arbitrary positive real number.
(iii) We say that $f : \Omega \to \mathbb{C}$ is periodic with period w if $f(z+w) = f(z)$ for all $z \in \Omega$. The function $\exp(z)$ is periodic, where every period is of the form $2\pi ni$ for some $n \in \mathbb{Z}$.
(iv) $e^z = 1$ if and only if $z = 2\pi ni$ for some $n \in \mathbb{Z}$.

Exercise
For $x, y \in \mathbb{R}$, we have $(e^x)^y = (e^y)^x$. Is this true for $\mathbb{C}$?

Connection to Polar Form Write $z = x + yi$ in polar form, $z = |z|(\cos\theta + i\sin\theta)$ for some $\theta \in \mathbb{R}$. But $\cos\theta + i\sin\theta = \exp(\theta i)$, so $z = |z|e^{\theta i}$.

This identity enables us to understand power functions $f(z) = z^n$ better:
$z^n = |z|^n e^{\theta i})^n = |z|^n e^{n\theta i}$.

Example
When $f(z) = z^2$, the upper half of the unit circle gets mapped onto the unit circle.
Inverting the Complex Exponential
The function $\exp : \mathbb{C} \to \mathbb{C} \setminus \{0\}$ does not have an inverse, because it is not one-to-one (e.g. $-1 = e^{\pi i} = e^{3\pi i}$). We need to find a region S inside $\mathbb{C}$ such that $\exp_S$ is one-to-one and onto.

Proposition
Let $S = \{x + iy : -\pi < y \leq \pi\}$. Then $\exp|_S : S \to \mathbb{C} \setminus \{0\}$ is one-to-one and onto.

Proof:
Let $z_1, z_2 \in S$ and suppose that $e^{z_1} = e^{z_2}$. Then $e^{z_1 - z_2} = 1$. By (iv), $z_1 - z_2 = 2n\pi i$ for some $n \in \mathbb{Z}$, in particular they lie on the same vertical and $|z_1 - z_2| = 2\pi n$. But the distance function between any two points in S on the same vertical line is less than 2π, so $z_1 = z_2$.
Let $w \in \mathbb{C} \setminus \{0\}$. We want to solve $w = e^{x+iy}$ for $x, y \in \mathbb{R}$. Write $w = |w|e^{i\theta}$ in polar coordinates, choosing $-\pi < \theta \leq \pi$. Then $|w|e^{i\theta} = e^x e^{yi}$. Set $x = \ln|w|$, and $y = \theta$. Then $x + yi = \ln|w| + \theta i \in S$. ■

Branch of the Argument Function
More generally for $b \in \mathbb{R}$, define $S_b = \{x + yi : b - 2\pi < y \leq b\}$ is one-to-one and onto.

Definition
Let $b \in \mathbb{R}$. The argument function $\arg : \mathbb{C} \setminus \{0\} \to (b - 2\pi; b]$ is defined as $\arg(x + iy) = \theta$, where $\theta \in \mathbb{R}$ is a unique number such that $x = |z|\cos\theta$ and $y = |z|\sin\theta$ and $b - 2\pi < \theta \leq b$. This is called a choice of branch of the argument function. If $b = \pi$, we call this the principal branch.

Definition
Let $b \in \mathbb{R}$ and let $\arg : \mathbb{C} \setminus \{0\} \to (b - 2\pi, b]$ be the argument function with prescribed branch. The function $\log : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ given by $\log(z) = \ln|z| + i\arg(z)$ is called the complex logarithm function. When $b = \pi$, we say that the principal branch of the complex logarithm is chosen. We have $\text{Range}(\log) = S_b$.

Properties of the Complex Logarithm
Fix $b \in \mathbb{R}$ and let arg have a branch $(b - 2\pi; b]$. Let $\log z = \ln|z| + i\arg z$. Then
(i) For every $z \in \mathbb{C} \setminus \{0\}$, $\exp(\log z) = z$. For every $z \in S_b$, $\log(\exp z) = z$.
(ii) For every $z_1, z_2 \in \mathbb{C} \setminus \{0\}$, $\log(z_1 z_2) = \log(z_1) + \log(z_2) + 2\pi k i$ for $k \in \mathbb{Z}$. (Exercise, show we need the $2\pi k i$)
(iii) Let $\Omega_b = \mathbb{C} \setminus (\{0\} \cup \{z \neq 0 : \arg(z) = b\})$. Then $\log z$ is analytic on Ω_b.
(iv) $\frac{d}{dz}\log z = \frac{1}{z}$ on Ω_b.
Proof of (iv):
$\log(z) = \ln|z| + i\arg z$. But $\ln|z| = \frac{1}{2}\ln(x^2 + y^2)$, so $\frac{\partial}{\partial x}\ln|z| = \frac{x}{x^2+y^2}$, $\frac{\partial}{\partial y}\ln|z| = \frac{y}{x^2+y^2} = -\frac{\partial}{\partial x}\arg z$ by the Cauchy-Riemann equations, so $\frac{d}{dz}\ln z = \frac{\partial u}{\partial x} + i\frac{\partial u}{\partial y} = \frac{x-iy}{x^2+y^2} = \frac{\overline{z}}{z\overline{z}} = \frac{1}{z}$. ■

Inverting the Power Function
Goal: For $n \in \mathbb{N}$, $n > 1$, define a function $\sqrt[n]{z}$ meaningfully (so $\sqrt[n]{z}^n = z$).
Problem: The map $z \mapsto z^n$ is not one-to-one, so we cannot invert it. For nonzero $w \in \mathbb{C}$, the equation $z^n = w$ has n solutions. Which one do we choose?

Definition
We say that $z \in \mathbb{C}$ is an nth root of unity iff there exists $n \in \mathbb{N}$ such that $z^n = 1$. We say that it is a primative nth root of unity iff $z^n = 1$ and $z^k \neq 1$ for $k < n - 1 \in \mathbb{N}$.

Definition
Idea: For $z \neq 0$, write $z^{1/n} = \exp((\log z)/n)$.
Def:
Let $\log : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ be a complex logarithm function, with a branch chosen so its values lie on the strip S_b. The nth root function is defined to be $\sqrt[n]{\ } : \mathbb{C} \setminus \{0\} \to \mathbb{C}$, $\sqrt[n]{z} = \exp(\frac{1}{n}\log(z))$.

Example
With the principle branch, $\frac{1}{4}\log(i) = \log(e^{\frac{\pi}{2}})/4 = \frac{i\pi}{8}$, so $\sqrt[n]{z} = e^{i\pi/8}$.

Definition
Let $\log : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ be a complex logarithm function, with a branch chosen chosen its values lie on a strip S_b. For $c \in \mathbb{C}$, define $z^c = \exp(c\log z)$. This is analytic when we exclude the ray on angle b, and

$\frac{d}{dz}z^c = cz^{c-1}$.

Definition
We say that $z \in \mathbb{C}$ is a nth root of unity if and only if there exists $n \in \mathbb{N}$ such that $z^n = 1$. We say that it is a primitive root nth root of unity if $z^k \neq 1$ for $k < n \in \mathbb{N}$ as well.
Exercise:
Prove every nth root of unity is of the form $\exp(2\pi k/n)$ for some $k \in \mathbb{Z}$, then prove if it is primitive it must have $\gcd(k, n) = 1$.

Definition
For $z \in \mathbb{C}$, define $\cos z = \frac{e^{iz}+e^{-iz}}{2}$ and $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$. Similarly, define $\cosh z = \frac{e^z+e^{-z}}{2}$ and $\sinh z = \frac{e^z-e^{-z}}{2}$.

Definition
If $u \in \mathcal{C}^2(\Omega)$ is harmonic in Ω, does there exist $v : \Omega \to \mathbb{R}$ such that $f = u + vi$ is analytic in Ω? Yes, if Ω is simply connected.

Integral Calculus
Let Ω be a domain and let $f : \Omega \to \mathbb{C}$ be a function on this domain, with Γ a piecewise smooth curve in Ω. We want to make sense of $\int_\Gamma f(z)dz$. Let $f(z) = u(z) + iv(z)$, $z = x + iy$, with $dz = dx + idy$.

Then

$$\int_\Gamma f(z)dz = \int_\Gamma (u(z) + iv(z))(dx + idy) = \int_\Gamma u(z)dx - v(z)dy + i\left(\int_\Gamma v(z)dx + u(z)dy\right)$$

Let $\alpha : [a, b] \to \mathbb{C}$ be a parameterization of Γ, with $\alpha(t) = \alpha_1(t) + i\alpha_2(t)$, so our first integral above becomes

$$\int_\Gamma u(z)dx - v(z)dy = \int_a^b (u(\alpha(t))\alpha_1'(t) - v(\alpha(t))\alpha_2'(t))dt$$

and similarly for $\int_\Gamma v(z)dx + u(z)dy$.

Example
Let's calculate $\int_\Gamma zdz$ where Γ is the quarter circle from 1 to z. We parameterize this with $\alpha(t) = \cos t + i \sin t$, with $t \in [0, \pi/2]$.

So

$$\int_\Gamma zdz = \int_\Gamma (x + iy)(dx + idy) = \int_\Gamma xdx - ydy + i\int_\Gamma ydx + xdy$$

$$= \int_0^{\pi/2} ((\cos t)(-\sin t) - \sin t(\cos t))dt + i\int_0^{\pi/2} ((\sin t)(-\sin t) + \cos t(\cos t))dt$$

$$= \int_0^{\pi/2} -2\sin t\cos t dt + i\int_0^{\pi/2} -\sin^2 t + \cos^2 t dt$$

$$= \int_0^{\pi/2} -\sin(2t)dt + i\int_0^{\pi/2} \cos(2t)dt$$

$$= [1/2\cos(2t)]_0^{\pi/2} + i[1/2\sin(2t)]_0^{\pi/2}$$

$$= -1$$

This is a lot of work, especially for a relatively simple function with a simple curve. This would get really hard if we just threw more functions into this naive way of computing the line integral. Can we generalize the fundamental theorem of calculus here? Unlike the real case, we can't just define some canonical antiderivative $G(z) = \int_a^z g(t)dt$, because there's no well-defined path from a to z. But we can write the FTC as $\int_a^b g(x)dx = \int_a^b G'(x)dx = \int_a^b dG$, but we've seen that for $u : \Omega \to \mathbb{R}$, $\int_\Gamma du = u(z_1) - u(z_o)$

where Γ is a curve from z_o to z_1.

Theorem
Let $f(z)$ be continuous in a domain Ω and suppose $F(z)$ is an analytic function on Ω satisfying $F'(z) = f(z)$ for all $z \in \Omega$. If Γ is a curve in Ω from z_o to z_1, then $\int_\Gamma f(z)dz = \int_\Gamma F'(z)dz = F(z_1) - F(z_o)$. Hence the value $\int_\Gamma f(z)dz$ only depends on the endpoints of Γ and not Γ itself.
Proof:
Write $f = u + iv$, $F = U + iV$. By the Cauchy-Riemann equations, $u = U_x = V_y$, where $v = V_x = -U_y$.
So

$$\int_\Gamma f(z)dz = \int_\Gamma udx - vdy + i\int_\Gamma vdx + udy$$

$$= \int_\Gamma U_x dx + U_y dy + i\int_\Gamma V_x dx + V_y dy$$

$$= \int_\Gamma dU + i\int_\Gamma dV$$

$$= U(z_1) - U(z_o) + i(V(z_1) - V(z_o)) = U(z_1) + iV(z_1) - (U(z_o) + iV(z_o)) = F(z_1) - F(z_o) \quad \blacksquare$$

Example
Let's do the same example we did earlier, $\int_\Gamma zdz$ where Γ is the quarter circle from 1 to i. Take $F(z) = \frac{1}{2}z^2$, with $F'(z) = z$ and F analytic. Therefore, by the theorem above, $\int_\Gamma zdz = F(i) - F(1) = -1$. Much less work. This is also true for any Γ between 1 and i.

Example
Consider the integrals $\int_{\Gamma_1} z^{-1}dz$ and $\int_{\Gamma_2} z^{-1}dz$, where Γ_1 and Γ_2 are counterclockwise and clockwise paths along the unit circle from 1 to -1.
Then

$$\int_{\Gamma_1} z^{-1}dz = \int_{\Gamma_1} \overline{z}dz = \int_{\Gamma_1} (x - iy)(dx + idy) = \int_{\Gamma_1} xdx + ydy + i\int_{\Gamma_1} (-ydx + xdy)$$

$$= \int_0^\pi (\cos(t)(-\sin(t)) + \sin(t)\cos(t))dt + i\int_0^\pi (-\sin t(-\sin t) + \cos(t)\cos(t))dt$$

$$= i\pi$$

In the other direction,

$$\int_{\Gamma_2} z^{-1}dz = \int_{\Gamma_2} \overline{z}dz = \int_{\Gamma_2} (x - iy)(dx + idy) = \int_{\Gamma_2} xdx + ydy + i\int_{\Gamma_2} (-ydx + xdy)$$

$$= \int_0^\pi (\cos(t)(-\sin(t)) - \sin(t)(-\cos(t)))dt + i\int_0^\pi (\sin t(-\sin t) + \cos(t)(-\cos(t)))dt$$

$$= -i\pi \neq \int_{\Gamma_1} z^{-1}dz$$

We see the theorem does not hold apply here. There is no analytic function $F : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ with $F'(z) = \frac{1}{z}$. The logarithm doesn't work because we need to choose a branch for the function to be defined, but this creates a discontinuity (preventing analyticity) along at least one of the curves, which we can't extend the logarithm through.

If Γ is the unit circle parameterized counterclockwise, then $\int_\Gamma z^{-1}dz = \int_{\Gamma_1} z^{-1}dz + \int_{-\Gamma_2} z^{-1}dz = i\pi - (-i\pi) = 2\pi i$.

Cauchy's Integral Theorem (for Jordan Curves):
Let $f : \Omega \to \mathbb{C}$ be analytic in Ω and let Γ be a Jordan curve such that $\Gamma \subseteq \Omega$ and $\operatorname{Int}\Gamma \subseteq \Gamma$. Then

$$\int_\Gamma f(z)dz = 0$$

Proof:
We have by Green's Theorem,

$$\int_\Gamma f(z)dz = \int_\Gamma (udx - vdy) + i\int_\Gamma (vdx + udy) = \iint_{\operatorname{Int}\Gamma} (-v_x - u_y)dydx + i\iint_{\operatorname{Int}\Gamma} (u_x - v_y)dxdy = 0$$

where both integrals are zero because their integrands vanish by the Cauchy-Riemann equations. ■

This theorem holds more generally, but where Ω is a k-connected Jordan domain Ω, where f is analytic on some domain Ω^+. Then $\int_{\partial\Omega} f(z)dz = 0$.

Definition
Let Ω be an open set, let Γ_1 and Γ_2 be two piecewise-smooth curves in Ω with the same endpoints z_o and z_1. Γ_1 and Γ_2 are called homotopic if we can continuously deform one into the other without leaving Ω. More formally, if we parameterize the curves by $\alpha : [0, 1] \to \Gamma_1$ and $\beta : [0, 1] \to \Gamma_2$, then α is homotopic fixed points endpoints to β if and only if there exists a continuous function $H : [0, 1] \times [0, 1] \to \Omega$ such that $H(0, t) = \alpha(t)$, $H(1, t) = \beta(t)$, and $H(s, 0) = z_o$ and $H(s, 1) = z_1$, for all $0 \le s \le 1$.

An interesting consequence of this is any closed curve is homotopic to to any point along it provided we can go through Ω.

Definition
Let Ω be an open set. We say that Ω is simply connected if and only if every closed curve Γ in Ω is homotopic as a closed curve to a point in Ω, informally, Ω has no holes.

Strong Cauchy's Integral Theorem
Let Ω be an open est, $f : \Omega \to \mathbb{C}$ is analytic in Ω. Let Γ be a piecewise-smooth closed curve (not necessarily simple) in Ω that is homotopic as a closed curve to a point in Ω. Then

$$\int_\Gamma f(z)dz = 0$$

This follows easily as a consequence:

Cauchy's Integral Theorem (for simply connected domains)
Let Ω be an open and simply connected subset of $\mathbb{C}$. Let Γ be a piecewise-smooth closed curve in Ω, and let $f : \Omega \to \mathbb{C}$ be analytic in Ω. Then

$$\int_\Gamma f(z)dz = 0$$

Corollary on Path Independence:
Let Ω be an open set, $f : \Omega \to \mathbb{C}$ analytic in Ω. Let Γ_1 and Γ_2 be the two piecewise smooth curves in Ω from z_0 to z_1 such that $\Gamma = \Gamma_1 \cup (-\Gamma_2)$ is homotopic as a closed curve to a point in Ω. Then

$$\int_{\Gamma_1} f(z)dz = \int_{\Gamma_2} f(z)dz$$

Lemma (ML-inequality):
Let $f(z)$ be a continuous function defined on the piecewise-smooth curve Γ. Suppose that there exists $M \ge 0 \in \mathbb{R}$, such that $|f(z)| \le M$ for all $z \in \Gamma$ and $L = \text{length}(\Gamma)$. Then

$$\left|\int_\Gamma f(z)dz\right| \le \int_\Gamma |f(z)||dz| \le ML$$

Here $|dz| = ds$ signifies the arc length parameterization of Γ.
Proof:
Write $I = \int_\Gamma f(z)dz$. Then $I = |I|e^{\omega i}$ for $\omega \in \mathbb{R}$. Thus,

$$|I| = \int_\Gamma e^{-\omega i} f(z)dz = \int_a^b e^{-iw} f(z(t))z'(t)dt$$

where $z(t)$ is a parameterization of Γ, $a \leq t \leq b$. Write $e^{-wi} f(z(t))z'(t) = U(t) + V(t)i$. Then,

$$|I| + i \cdot 0 = \int_a^b U(t)dt + i\int_a^b V(t)dt$$

Equating real and imaginary parts,

$$|I| = \int_a^b U(t)dt$$

and $0 = \int_a^b V(t)dt$.

Since everything is real,

$$|I| \leq \int_a^b |U(t)|dt \leq \int_a^b |e^{-\omega i} f(z(t))||z'(t)|dt$$

$$= \int_a^b |f(z(t))||z'(t)|dt$$

$$\leq M\int_a^b |z'(t)|dt$$

$$= ML \quad \blacksquare$$

Antiderivative Theorem

Let $\Omega \subseteq \mathbb{C}$ be open and simply connected. Let $f : \Omega \to \mathbb{C}$ be analytic in Ω. There exists a function $F : \Omega \to \mathbb{C}$ that is analytic in Ω such that $F'(z) = f(z)$ for every $z \in \Omega$. This function is unique up to an additive constant.
Proof:
Fix $z_o \in \Omega$. Since Ω is simply connected, we have $\int_{\Gamma_1} f(\zeta)d\zeta = \int_{\Gamma_2} f(\zeta)d\zeta$ for any two piecewise smooth curves Γ_1, Γ_2 in Ω from z_0 to $z \in \Omega$. So the function $F : \Omega \to \mathbb{C}$ $F(z) = \int_\Gamma f(\zeta)d\zeta$, where Γ is any piecewise smooth path from z_0 to z. Let us show that $F'(z) = f(z)$ for every $z \in \Omega$. Fix $z \in \Omega$. Let $\varepsilon > 0$. Since Ω is open and f is continuous at z, there exists $\delta > 0$ such that $D(z;\delta) \subseteq \Omega$ and $|f(\zeta) - f(z)| < \varepsilon$ for every $\zeta \in D(z;\delta)$. Let $w \in D(z;\delta)$ so that $0 < |w - z| < \delta$, and connect z to w by a straight line segment ρ (since both are in $D(z;\delta)$), which in turn connects with Γ from z_0 to w, with $\rho \subseteq D(z;\delta)$ and we have

$$F(w) - F(z) = \int_{\Gamma+\rho} f(\zeta)d\zeta - \int_\Gamma f(\zeta)d\zeta = \int_\rho f(\zeta)d\zeta$$

Hence

$$\left|\frac{F(w) - F(z)}{w - z} - f(z)\right| = \left|\frac{f(w) - f(z) - (w - z)f(z)}{w - z}\right|$$

$$= \left|\frac{\int_\rho f(\zeta)d\zeta - f(z)\int_\rho d\zeta}{w - z}\right|$$

$$= \left|\frac{\int_\rho (f(\zeta) - f(z))d\zeta}{w - z}\right|$$

$$\leq \frac{\varepsilon \overbrace{\int_\rho |d\zeta|}^{\text{length of }\rho}}{|w-z|}$$

$$= \frac{\varepsilon|w-z|}{|w-z|} = \varepsilon$$

Therefore, $F'(z) = \lim_{w\to z} \frac{F(w)-F(z)}{w-z} = f(z)$. F' is continuous because f is continuous, therefore F is analytic on Ω.

Proving uniqueness up to a constant, suppose $F_0 : \Omega \to \mathbb{C}$ such that F_0 is analytic in Ω and $F_0'(z) = f(z)$ for all $z \in \Omega$. Let $G(z) = F(z) - F_0(z)$. Then G is analytic in Ω, with $G'(z) = 0$. By A2Q2, this tells us G is constant. ■

Exercise
Let $\Omega \subseteq \mathbb{C}$ be open, and let $f : \Omega \to \mathbb{C}$ be a function such that $f'(z)$ exists for all $z \in \Omega$. Then f is continuous on Ω.

Exercise
Prove Taylor's Theorem: Let Ω be open, $z_o \in \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be a complex function such that $f'(z_o)$ exists. Prove there exists a function $\psi : \Omega \to \mathbb{C}$ such that $f(z) = f(z_o) + f'(z_o)(z-z_o) + \psi(z)(z-z_o)$ and $\lim_{z\to z_o} \psi(z) = 0$.

Theorem (Goursat): Let Ω be an open subset of $\mathbb{C}$. Let $T \subseteq \Omega$ be a triangle whose interior is contained in Ω. Let $f : \Omega \to \mathbb{C}$ be such that $f'(z)$ exists for all $z \in \Omega$. Then $\int_T f(z)dz = 0$.
Proof:
TRANSCRIBE LATER. ■

Corollary:
Let $z_0 \in \mathbb{C}, R > 0$. Let f be a function s.t. $f'(z)$ exists for all $z \in D(z_o; R)$. Then $f(z)$ has a primitive on $D(z_o; R)$.
Proof:
TRANSCRIBE LATER. ■

Cauchy's Integral Formula (for Jordan curves)
Let $f(z)$ be analytic in an open set Ω, and $\Gamma \subseteq \Omega$ be a Jordan curve with $\operatorname{Int}\Gamma \subseteq \Omega$. Then for every $z_o \in \operatorname{Int}\Gamma$,

$$f(z_o) = \frac{1}{2\pi i}\int_\Gamma f(z)/(z-z_o)dz$$

Proof:
Let $r > 0$ be small enough that $\overline{D}(z_o; r) \subseteq \operatorname{Int}\Gamma$. Consider a new domain $\Omega' = \operatorname{Int}\Gamma \setminus \overline{D}(z_o; r)$. This is a 2-connected Jordan domain. Its boundary is $\partial\Omega' = \Gamma \cup (-C(z_o; r))$ (with a choice of orientation). Hence $\frac{1}{2\pi i}\int_{\Gamma\cup(-C(z_o;r))} f(z)/(z-z_o)dz = 0$ by Cauchy's Theorem, but then this tells us $\frac{1}{2\pi i}\int_\Gamma f(z)/(z-z_o)dz = \frac{1}{2\pi i}\int_{C(z_o;r)} f(z)/(z-z_o)dz$. Thus we were able to replace integration over Γ with integration over a nice curve $C(z_o; r)$.

Now write $f(z) = (f(z)-f(z_o))+f(z_o)$, so that $\frac{1}{2\pi i}\int_C f(z)/(z-z_o)dz = \frac{f(z_o)}{2\pi i}\int_C \frac{dz}{z-z_o} + \frac{1}{2\pi i}\int_C \frac{f(z)-f(z_o)}{z-z_o}dz$. But we know this left integral (we computed it as an example), it's $2\pi i$, so we have $\frac{1}{2\pi i}\int_C f(z)/(z-z_o)dz = f(z_o) + \frac{1}{2\pi i}\int_C \frac{f(z)-f(z_o)}{z-z_o}dz$. Since $f'(z_o)$ exists, for every small $\varepsilon > 0$ there exists $\delta > 0$ if $z \in D(z_o; \delta) \setminus \{0\}$ then $|\frac{f(z)-f(z_o)}{z-z_o} - f'(z_o)| < \varepsilon$. Hence $|\frac{f(z)-f(z_o)}{z-z_o}| + |f'(z_o)| \leq |\frac{f(z)-f(z_o)}{z-z_o} - f'(z_o)|$ so

$|\frac{f(z)-f(z_o)}{z-z_o}| \le M = |f'(z)| + \varepsilon$ for every $z \in D(z_o;\delta) \setminus \{z_o\}$. By ML inequality,

$$|\frac{1}{2\pi i}\int_{C(z_0;R)} \frac{f(z)-f(z_o)}{z-z_o}dz| = |\frac{1}{2\pi i}\int_{C(z_0;\delta/2)} \frac{f(z)-f(z_o)}{z-z_o}dz| \le \frac{M}{2\pi}\cdot \text{length}(C(z_0;\delta/2)) = M\frac{\delta}{2}$$

Letting $\delta \to 0$, this vanishes, giving the result (the same curve shrinking trick). ■

Definition:
Let Γ be a piecewise-smooth closed curve in $\mathbb{C}$ and $z_o \in \mathbb{C}$ be a point not on Γ. The winding number of Γ with respect to z_o is

$$I(\Gamma, z_o) = \frac{1}{2\pi i}\int_\Gamma dz/(z-z_o)$$

Cauchy's Integral Formula (general)
Let $f(z)$ be analytic in an open set Ω, and let $\Gamma \subseteq \Omega$ be a piecewise-smooth closed curve that is homotopic as a closed curve to a point in Ω. Then for every $z_o \in \Omega$,

$$f(z_o)I(\Gamma, z_o) = \frac{1}{2\pi i}\int_\Gamma f(z)/(z-z_o)dz$$

Higher Order Derivatives
Theorem
Let Ω be an open subset of $\mathbb{C}$. Let $f : \Omega \to \mathbb{C}$ be analytic in Ω. Then all derivatives $f'(\zeta), f''(\zeta), \ldots, f^{(n)}(\zeta), \ldots$ exist and are analytic at all ζ in Ω. Moreover, we have the formula

$$f^{(n)}(\zeta) = \frac{n!}{2\pi i}\int_\Gamma \frac{f(z)}{(z-\zeta)^{n+1}}dz$$

where Γ is any Jordan curve inside Ω containing ζ in its interior.
Proof:
By induction on $n \ge 0$. For $n = 0$ the formula is Cauchy's Integral formula. Also $f = f^{(0)}$ is analytic on Ω by hypothesis. Now assume that $f^{(k)}(\zeta) = \frac{k!}{2\pi i}\int_\Gamma \frac{f(z)}{(z-\zeta)^{k+1}}dz$.
Then

$$\begin{aligned} f^{(k+1)}(\zeta) &= \frac{d}{d\zeta}f^{(k)}(\zeta) \\ &= \frac{k!}{2\pi i}\frac{d}{d\zeta}\int_\Gamma \frac{f(z)}{(z-\zeta)^{k+1}}dz \\ &= \frac{k!}{2\pi i}\int_\Gamma \frac{d}{d\zeta}\frac{f(z)}{(z-\zeta)^{k+1}}dz \\ &= \frac{(k+1)!}{2\pi i}\int_\Gamma \frac{f(z)}{(z-\zeta)^{k+2}}dz \end{aligned}$$

The only step to justify is why we can invoke Leibniz's integral rule to interchange the derivative and integral. The proof for the complex case reduces to the real case because our complex line integral is just two real line integrals, along with the Wirtinger derivative $\frac{d}{dz}f(z) = \frac{1}{2}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y})(u(x,y) + v(x,y)i)$. ■

We can also prove this with uniform convergence:
Definition
Let $f_n : \Omega \to \mathbb{C}$ be a sequence of complex functions. We say that f_n is uniformly convergent on Ω with limit $f : \Omega \to \mathbb{C}$ iff for all $\varepsilon > 0$, there exists $n_o \in \mathbb{N}$ such that for all $n \ge n_o$ and for all $z \in \Omega$, $|f_n(z) - f(z)| < \varepsilon$.

Lemma
When $f_n \to f$ uniformly, then $\lim_{n\to\infty}\int_\Gamma f_n(z)dz = \int_\Gamma f(z)dz$.
Proof:

$|\int_\Gamma f(z)dz - \int_\Gamma f_n(z)dz| = |\int_\Gamma f(z) - f_n(z)dz| \leq \sup_{z\in\Gamma}|f(z) - f_n(z)| \cdot \text{length}(\Gamma) \to 0$ by uniform convergence. ■

Now we prove our theorem again using uniform convergence:
Let ζ_n be a sequence in Ω such that $\zeta_n \to 0$ and $\zeta + \zeta_n \in \Omega$. Consider $f(\zeta+\zeta_n) = \frac{1}{2\pi i}\int_\Gamma \frac{f(z)}{z-\zeta-\zeta_n}dz$ and $f(\zeta) = \frac{1}{2\pi i}\int_\Gamma \frac{f(z)}{z-\zeta}dz$. Then $\frac{f(\zeta+\zeta_n)-f(\zeta)}{\zeta_n} = \frac{1}{2\pi}\int_\Gamma \frac{f(z)}{(z-\zeta)^2-\zeta_n(z-\zeta)}dz$ after some simplification. Now, the function $g_n(z) = \frac{f(z)}{(z-\zeta)^2-\zeta_n(z-\zeta)} \to g(z) = \frac{f(z)}{(z-\zeta)^2}$ uniformly. Conclude that

$$f'(\zeta) = \lim_{n\to\infty}\frac{f(\zeta+\zeta_n)-f(\zeta)}{\zeta_n} = \lim_{n\to\infty}\int_\Gamma g_n(z)dz = \int_\Gamma g(z)dz = \frac{1}{2\pi i}\int_\Gamma \frac{f(z)}{(z-\zeta)^2}dz$$

and by a similar inductive proof for higher derivatives the result holds, and all higher order derivatives are analytic in Ω. ■

Example
Compute $I = \int_{-\infty}^{\infty}\frac{dx}{(x^2+1)^2}$.
Sol:
Notice that $I = \lim_{R\to\infty}\int_{-R}^{R}(x^2+1)^{-2}dx$. Consider the straight line from $-R$ to R given by Γ, and the semicircle from R to $-R$, called C_R. Then,

$$\int_{\Gamma\cup C_r}\frac{dz}{(z^2+1)^2} = I_R + \int_{C_r}\frac{dz}{(z^2+1)^2}$$

Let $f(z) = \frac{1}{(z+i)^2}$, $z_o = i$, $n = 1$. Then $f(z)$ is analytic on some open Ω^+, with both $\text{Int}(\Gamma\cup C_R)\cup(\Gamma\cup C_r) \subseteq \Omega^+$. Hence the formula for higher order derivatives applies:

$$f'(z_o) = \frac{1}{2\pi i}\int_{\Gamma\cup C_R} = \frac{1}{2\pi i}\int_{\Gamma\cup C_R}\frac{dz}{(z^2+1)^2}$$

But $f'(z_o) = \frac{-2}{z+i}^3 = -\frac{2}{(2i)^3} = -\frac{1}{4}i$, hence

$$\int_{\Gamma\cup C_R}\frac{dz}{(z^2+1)^2} = \frac{\pi}{2}$$

Hence

$$I_R + \int_{C_R}\frac{dz}{(z^2+1)^2} = \frac{\pi}{2}$$

We claim this last integral goes to 0 as R goes to infinity. Parameterize C_R by $z = Re^{\theta i}$, $0 \leq \theta \leq \pi$. Hence $dz = Rie^{\theta i}d\theta$, and

$$\int_{C_R}\frac{dz}{(z^2+1)^2} = \int_0^\pi \frac{Rie^{\theta i}}{(R^2e^{2\theta i}+1)^2}d\theta$$

so

$$|\int_{C_R}\frac{dz}{(z^2+1)^2} = |\int_0^\pi \frac{Rie^{\theta i}}{(R^2e^{2\theta i}+1)^2}d\theta|$$
$$\leq \int_0^\pi |\frac{Rie^{\theta i}}{(R^2+e^{2\theta i}+1)^2}|d\theta$$
$$= R\int_0^{2\pi}\frac{d\theta}{|R^2e^{2\theta i}+1|^2}$$

But by the reverse triangle inequality, $|R^2e^{2\theta i}+1| \geq |R^2e^{2\theta i}| - 1 = R^2 - 1$, hence the above is

$$\leq R\int_0^\pi \frac{d\theta}{(R^2-1)^2} = \frac{R}{(R^2-1)^2}\pi$$

which goes to 0 as R goes to infinity.

Therefore, $\lim_{R\to\infty} I_R + \int_{C_r} \frac{dz}{(z^2+1)^2} = \pi/2 \implies I = \frac{\pi}{2}$.

Theorem (Harmonicity of u and v)

Let $f = u + vi$ be analytic in Ω. Then u and v are harmonic in Ω.

Proof:

Since f is analytic in Ω, by our above result f' and f'' exist and are analytic in Ω, so $u_{xx}, u_{yy}, v_{xx}, v_{yy}$ all exist and are continuously differentiable by the Cauchy-Riemann equations (applied to f'') and we have $u_{xx} = -u_{yy}$ and $v_{xx} = -v_{yy}$ by the Cauchy-Riemann equations (applied to f'). ■

Circumferential Mean value Theorem (for analytic functions)

Let f be analytic in an open set Ω. Let $z_o \in \Omega$, and $r > 0 \in \mathbb{R}$ such that $\overline{D}(z_o; r) \subseteq \Omega$. Then $f(z_o) = \frac{1}{2\pi} \int_0^{2\pi} f(z_o + re^{\theta i})d\theta$.

Proof:

We parameterize $C(z_o; r)$ by $z = z_o + re^{i\theta}$ for $0 \leq \theta \leq 2\pi$. Then $dz = rie^{\theta i}d\theta$. By Cauchy's Integral formula,

$$f(z_o) = \frac{1}{2\pi i} \int_{C(z_o;r)} \frac{f(z)}{z - z_o} dz = \frac{1}{2\pi i} \int_{C(z_o;r)} \frac{f(z_o + re^{\theta i})}{re^{\theta i}} ire^{\theta i} d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(z_o + re^{i\theta})d\theta \quad ■$$

Solid Mean Value Theorem (for analytic functions)

Let f be analytic in an open set Ω. Let $z_o \in \Omega$, $r > 0 \in \mathbb{R}$ such that $\overline{D}(z_o; r) \subseteq \Omega$. Then $f(z_o) = \frac{1}{\pi r^2} \iint_{D(z_o;r)} f(z)dxdy$.

Proof (sketch):

Split $f = u + iv$, apply the corresponding theorem for harmonic functions. ■

Definition

Let $f : \Omega \to \mathbb{C}$ be a complex function. We say f attains its maximum modulus at $z_o \in \Omega$ if and only if $|f(z)| \leq |f(z_o)|$ for all $z \in \Omega$.

Theorem (Strong Maximum Modulus Principle)

Let f be a nonconstant analytic function defined in a domain Ω. Then f does not assume its maximum modulus at any point of Ω.

Proof:

If $|f(z)|$ is unbounded for $z \in \Omega$, we're done. So assume that there exists $M > 0$ such that $|f(z)| \leq M$ for all $z \in \Omega$. Suppose that f attains its maximum modulus at z_o on Ω. Define $g(z) = f(z) + C$, choosing $C \in \mathbb{R}$ large s.t. $g(z) \neq 0$ for all $z \in \Omega$.

Define $h : \Omega \to \mathbb{C}$, $h(z) = \ln |g(z)|$. By A2, $h(z)$ is harmonic in Ω. Since $\ln(x)$ is monotically increasing on $(0, \infty)$, $h(z)$ assumes its maximum at z_o. By the Strong Maximum Principle for harmonic functions, $h(z)$ is constant. But then $|g(z)| = |f(z) + C|$ is constant, which is impossible, because $f(z)$ is not constant.

Theorem (Weak Maximum Modulus Principle)

Let Ω be a bounded domain and f a nonconstant function analytic on Ω and continuous on its closure $\overline{\Omega}$. Then f attains its maximum modulus $|f(z)|$ on $\partial\Omega$ only.

Proof:

Follows immediately from the real harmonic case. ■

Definition

We say that $f : \Omega \to \mathbb{C}$ is bounded iff $f(\Omega)$ is a bounded subset of $\mathbb{C}$.

Definition

A function $f : \mathbb{C} \to \mathbb{C}$ is entire if it is analytic on all of $\mathbb{C}$.

Theorem (Liouville)
A bounded entire function is constant.
Proof:
If $f : \mathbb{C} \to \mathbb{C}$, $f = u + vi$ is analytic in $\mathbb{C}$ and bounded, then both u and v are bounded harmonic entire functions (implicitly since f's second derivative exists, whose real and imaginary parts are the second partials of u and v). The result follows from Liouville's theorem for harmonic functions. ■

Theorem (Fundamental Theorem of Algebra)
For every polynomial $p(x) \in \mathbb{C}[x]$ of degree $n \geq 1$, there exists $z_o \in \mathbb{C}$ such that $p(z_o) = 0$.
Proof:
Suppose not, so $p(z) \neq 0$ for all $z \in \mathbb{C}$. We claim that this implies there exists $M > 0$ such that $|p(z)| \geq M$ for all $z \in \mathbb{C}$. If that's the case, then $f(z) = \frac{1}{p(z)}$ is bounded because $f(\mathbb{C})$ is contained in $\overline{D}(0; \frac{1}{M})$. Then, by Liouville's Theorem (since f is analytic because it is rational), $f(z)$ is constant, which is impossible since $p(z)$ is not constant, our contradiction.

Suppose that such an M doesn't exist. Then there exists a sequence $z_1, z_2, \ldots,$ such that $\lim_{k\to\infty} p(z_k) = 0$. Because p is continuous, this tells us $p(\lim_{k\to\infty} z_k) = 0$. Suppose that (z_k) is unbounded, so that there is a subsequence (z_{k_n}) such that $|z_{k_n}| \to \infty$ as $n \to \infty$. Write $p(z) = a_n z^n + \cdots + a_0, a_n \neq 0$. Then $|p(z)| \geq |a_n||z|^n - \sum_{k=0}^{n-1} |a_k||z|^k$ and for sufficiently large $|z|$, $\sum_{k=0}^{n-1} |a_k||z|^k \leq \frac{|a_n|}{2}|z|^n$. So $|p(z_{k_n})| \geq \frac{|a_n|}{2}|z_{k_n}|^n$ for sufficiently large $|z_{k_n}|$. This contradicts the fact that $\lim_{n\to\infty} |p(z_{k_n})| = 0$, hence (z_k) must be bounded, so by Bolzano-Weierstrass, it has a convergent subsequence (w_n) such that $\lim_{n\to\infty} w_n = w$ for some w. Since $\mathbb{C}$ is complete, it contains all of its limits, so $w \in \mathbb{C}$. Since p is continuous, $0 = \lim_{n\to\infty} p(w_n) = p(\lim_{n\to\infty} w_n) = p(w)$. Hence w is a root of p, contradiction. ■

Theorem (Path Independence Theorem)
Suppose f is a continuous function on a domain Ω. The following are equivalent:
(i) (Integrals are path independent):
If z_o, z_1 are any two points in Ω and Γ_1, Γ_2 are piecewise-smooth closed curves in Ω from z_o to z_1, then $\int_{\Gamma_1} f(z)dz = \int_{\Gamma_2} f(z)dz$.
(ii) (Integrals around closed curves are 0):
If Γ is a piecewise-smooth closed curve in Ω, then $\int_\Gamma f(z)dz = 0$.
(iii) (There is a global antiderivative for f on Ω)
There is a function $F : \Omega \to \mathbb{C}$ that is analytic in Ω such that $F'(z) = f(z)$ for all $z \in \Omega$.
Proof:
The equivalence of (i) and (ii) isn't too bad (we've seen it before), (iii) implies (i) is an exercise, and (ii) implies (iii) is more or less the antiderivative theorem. ■

Theorem (Morera)
Let $f : \Omega \to \mathbb{C}$ be continuous on a domain Ω, and suppose $\int_\Gamma f(z)dz = 0$ for every piecewise-smooth closed curve Γ in Ω. Then f is analytic in Ω.
Proof:
By Path Independence, the fact that integrals around closed curves are 0 implies that there is a global antiderivative F of f in Ω. Since F is analytic in Ω, it follows from Cauchy's Integral Formula for higher-order derivatives that F'' is analytic in Ω, but $F'' = f'$, meaning that f' exists in Ω and is continuous. Hence f is analytic in Ω. ■

Theorem (Cauchy's Inequality)
Let f be analytic in an open set Ω. Let $z_o \in \Omega$, $r > 0$, and suppose that $\overline{D}(z_o; r) \subseteq \Omega$. Write $M(z_o; r) = \max_{z\in C(z_o;r)} |f(z)|$. Then for every $n = 0, 1, \ldots$, $|f^{(n)}(z_o)| \leq \frac{n! M(z_o;r)}{r^n}$.
Proof:
By Cauchy's Integral Formula for higher order derivatives, $|f^{(n)}(z_o)| = |\frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-z_o)^{n+1}} dz| \underbrace{\leq}_{ML} \frac{n!}{2\pi} \frac{M(z_o;r)}{r^{n+1}} \text{length}(C(z_o; r))$

$\frac{n!2\pi r M(z_o;r)}{2\pi r^{n+1}}$. ■

Part V - Series

Goal:

Definition

Let Ω be open, $f : \Omega \to \mathbb{C}$ a complex function. We say that f is analytic at $z_o \in \Omega$ if and only if there exists $a_0, a_1, \dots \in \mathbb{C}$ such that $f(z) = \sum_{n=0}^{\infty} a_n(z - z_o)^n$ for all z in some open neighbourhood of z_o. We will show that our definition of analytic (a.k.a holomorphic) and this new definition of analytic are equivalent. Over $\mathbb{R}$ this is false. Take $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \exp(-1/x^2)$ for $x \neq 0$ and $f(x) = 0$ for $x = 0$. Every order of its derivative exists, but the Taylor series of $f(x)$ at 0 is not equal to $f(x)$ on any neighborhood of 0.

Definition

Let $(z_1)_{n=1}^{\infty}$ be a sequence in $\mathbb{C}$. This sequence is said to converge to $z_o \in \mathbb{C}$ if and only if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, the inequality $|z_n - z_o| < \varepsilon$ holds.

Definition

An infinite series $\sum_{k=1}^{\infty} a_k$ for $a_k \in \mathbb{C}$ is said to converge to $S \in \mathbb{C}$ if and only if the sequences of partial sums $s_n = \sum_{k=1}^{n} a_k$ converges to S.

Notation

For sequences $z_n \underset{n\to\infty}{\to} z_o$, for series $\sum_{k=1}^{\infty} a_k = S$.

Properties

The limit z_o (resp. S) is unique.

$(z_n)_{n=1}^{\infty}$ converges if and only if it is Cauchy, i.e. for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, we have $|z_n - z_m| < \varepsilon$.

Cauchy Criterion: If $\sum_{k=1}^{\infty} a_k$ converges then $a_k \to 0$. The converse is not true (harmonic series).

Definition

We say that the series $\sum_{k=1}^{\infty} a_k$ converges absolutely if and only if $\sum_{k=1}^{\infty} |a_k|$ converges.

Exercise

Prove that if $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges. The converse is not true (alternating harmonic series).

Definition

Let $f_n : \Omega \to \mathbb{C}$ be a sequence of complex functions. We say that f_n converges pointwise on Ω if and only if the sequence $f_n(z)$ converges for all $z \in \Omega$. We say that f_n converges uniformly on Ω with limit $f : \Omega \to \mathbb{C}$ if and only if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and $z \in \Omega$, $|f_n(z) - f(z)| < \varepsilon$.

Lemma

Let $f_n : \Omega \to \mathbb{C}$ be a sequence of continuous functions converging uniformly to f. Then $\lim_{n\to\infty} \int_\Gamma f_n(z)dz = \int_\Gamma f(z)dz$.

Corollary:

If $\sum_{k=1}^{\infty} g_k(z)$ converges uniformly on Γ, then $\int_\Gamma \sum_{k=1}^{\infty} g_k(z)dz = \sum_{k=1}^{\infty} \int_\Gamma g_k(z)dz$

Definition

Let $g_k : \Omega \to \mathbb{C}$ be a sequence of complex functions. A series $\sum_{k=1}^{\infty} g_k(z)$ is said to converge pointwise on Ω if and only if the corresponding partial sums $s_n(z) = \sum_{k=1}^{n} g_k(z)$ converge pointwise. A series $\sum_{k=1}^{\infty} g_k(z)$ is said to converge uniformly if and only if $s_n(z)$ converges uniformly on Ω.

Cauchy Criterion

(i) A sequence $f_n : \Omega \to \mathbb{C}$ converges uniformly on Ω if and only if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $p \in \mathbb{N}$ and $z \in \Omega$, if $|f_n(z) - f_{n+p}(z)| < \varepsilon$.

(ii) A series $\sum_{k=1}^{\infty} g_k(z)$ converges uniformly on Ω if and only if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $p \in \mathbb{N}$ and $z \in \Omega$, $|\sum_{k=n+1}^{n+p} g_k(z)| < \varepsilon$.

Theorem (Uniform Convergence of Continuous Functions)

If the sequence $f_n : \Omega \to \mathbb{C}$ consists of continuous functions on Ω that are uniformly convergent on Ω to the function f, then f is continuous on Ω. Similarly, if the functions $g_k(z)$ are continuous and $g(z) = \sum_{k=1}^{\infty} g_k(z)$ converges uniformly on Ω, then g is continuous on Ω.

Theorem (Weierstrass M-Test):

Let $g_n : \Omega \to \mathbb{C}$ be a sequence of functions. Suppose that there exists non-negative real constants $M_1, M_2, \dots$ such that $|g_n(z)| \leq M_n$ for all $z \in \Omega$, and $\sum_{n=1}^{\infty} M_n$ converges. Then $\sum_{n=1}^{\infty} g_n$ converges absolutely and uniformly on Ω.

Proof:

Since $\sum_k M_k$ converges, Cauchy's criterion applies: for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$, such that for all $n \geq N$, $p \in \mathbb{N}$, then $\sum_{k=n+1}^{n+p} M_k < \varepsilon$. So for $n \geq N$ and $z \in \Omega$, we have

$$\left|\sum_{k=n+1}^{n+p} g_k(z)\right| \leq \sum_{k=n+1}^{n+p} |g_k(z)| \leq \sum_{k=n+1}^{n+p} M_k < \varepsilon$$

so by the Cauchy criterion for uniform convergence, we are done. ■

Theorem (Analytic Convergence Theorem):

(i) Let Ω be open, and $f_n : \Omega \to \mathbb{C}$ be a sequence of functions that are holomorphic in Ω. If $f_n \to f$ uniformly on every closed disk contained in Ω, then f is holomorphic in Ω. Furthermore, $f_n' \to f'$ pointwise on Ω and uniformly on every closed disk in Ω.

(ii) Let $g_k : \Omega \to \mathbb{C}$ be a sequence of holomorphic functions in Ω, and $g(z) = \sum_{k=1}^{\infty} g_k(z)$ converges uniformly on every cloesd disk in Ω, then g is holomorphic in Ω and $g'(z) = \sum_{k=1}^{\infty} g_k'(z)$ pointwise in Ω and uniformly on every closed disk contained in Ω.

Proof:

We prove (i), and (ii) will follow by considering partial sums. Let $z_o \in \Omega$ and $r > 0 \in \mathbb{R}$ be such that $\overline{D}(z_o; r) \subseteq \Omega$. Notice that $f_n \to f$ uniformly on $D(z_o; r)$. We will show that f is analytic using Morera's Theorem. Notice that f is continuous on $D(z_o; r)$ since it is the uniform limit of analytic functions. Let Γ be any piecewise-smooth closed curve in $D(z_o; r)$. Since f_n is analytic, by Cauchy's Integral Theorem, $\int_\Gamma f_n(z)dz = 0$. But by the lemma proved in lecture 17, $\int_\Gamma f_n(z)dz \to \int_\Gamma f(z)dz$, so $\int_\Gamma f(z)dz = 0$. Thus, by Morera's Theorem, f is analytic on $D(z_o; r)$, so extending this to every point in Ω, f is analytic on Ω. It remains to show $f_n' \to f'$ converges uniformly on every closed disk in Ω. Suppose $\overline{D}(z_o; r) \subseteq \Omega$. By openness of Ω, there exists $\rho > r$ such that $\overline{D}(z_o; r) \subseteq \overline{D}(z_o; \rho) \subseteq \Omega$ (by picking r smaller as necessary earlier). For any $z \in \overline{D}(z_o; r)$, by Cauchy's Integral Formula,

$$f_n'(z) = \frac{1}{2\pi i} \int_{C(z_o;\rho)} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta$$

and

$$f'(z) = \frac{1}{2\pi i} \int_{C(z_o;\rho)} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta$$

Since $f_n \to f$ uniformly on $\overline{D}(z_o; \rho)$, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and $z \in \overline{D}(z_o; \rho)$, we have $|f_n(z) - f(z)| < \varepsilon$. In particular, $|f_n(\zeta) - f(\zeta)| < \varepsilon$ for every $\zeta \in C(z_o; \rho)$.

Therefore, for every $z \in D(z_o; r)$, we have

$$|f_n'(z) - f'(z)| = |\frac{1}{2\pi i} \int_{C(z_o;\rho)} \frac{f_n(\zeta) - f(\zeta)}{(\zeta - z)^2} d\zeta|$$

$$\underbrace{\leq}_{ML} \frac{1}{2\pi} \int_{C(z_o;\rho)} \frac{|f_n(\zeta) - f(\zeta)|}{|\zeta - z|^2} |d\zeta|$$

$$\underbrace{\leq}_{|\zeta - z| \geq \rho - r} \frac{\varepsilon}{2\pi} \frac{1}{(\rho - r)^2} 2\pi\rho$$

$$= \frac{\varepsilon\rho}{(\rho - r)^2}$$

Since ρ and r are fixed and independent of z, this goes to 0. ■

Definition

A power series is a series of the form $\sum_{n=0}^{\infty} a_n(z - z_o)^n$, where $a_o, a_1, \ldots$ are fixed complex numbers.

Theorem (Power Series Convergence Theorem)

Let $\sum_{n=0}^{\infty} a_n(z - z_o)^n$ be a power series. There is a unique $0 \leq R \leq \infty$ called the radius of convergence, such that if $|z - z_o| < R$, the series converges, and if $|z - z_o| > R$, the series diverges. Furthermore, this convergence is uniform and absolute on every closed disk in $D(z_o; R)$. Note that no general statement can be made about the boundary $C(z_o; R)$.

To prove this theorem, we'll need:

Lemma (Abel-Weierstrass)

Let $r_0 \geq 0 \in \mathbb{R}$ and suppose $|a_n| r_0^n \leq M$ for all n, where M is some constant. Then, for $r < r_0$, $\sum_{n=0}^{\infty} a_n(z - z_o)^n$ converges uniformly and absolutely on the closed disk $\overline{D}(z_o; r)$.

Proof:

For $z \in \overline{D}(z_o; r)$, we have $|a_n(z - z_o)^n| \leq |a_n| r^n \leq M(\frac{r}{r_0})^n$. Let $M_n = M(r/r_0)^n$. Since $r/r_0 < 1$, $\sum M_n$ converges, so by the Weierstrass M-test, the series converges uniformly and absolutely on $\overline{D}(z_o; r)$. ■

Proof of Power Series Convergence Theorem:

Set $R = \sup\{r \geq 0 : \sum_{n=0}^{\infty} |a_n| r^n \text{ converges}\}$. Let $0 < r_0 < R$ (if $R = 0$, we have nothing to prove). By definition of R, there exists r_1 such that $r_0 < r_1 \leq R$, and $\sum |a_n| r_1^n$ converges. Since $\sum |a_n| r_0^n < \sum |a_n| r_1^n$, by the Comparison Test the series on the left converges. The terms $|a_n| r_0^n$ are bounded, so by Abel-Weierstrass, $\sum a_n(z - z_o)^n$ converges and absolutely and uniformly on $\overline{D}(z_o; r)$ for every $r < r_0$. But every $z \in D(z_o; R)$ belongs to some closed disk contained in $D(z_o; R)$, so we can always choose r_0 to achieve convergence at z. For divergence, suppose that $|z - z_o| > R$ and assume that $\sum a_n(z - z_o)^n$ converges. The terms $a_n(z - z_o)^n$ are bounded in absolute value because the series converges. By Abel-Weierstrass Lemma, if $R < r < |z_1 - z_o|$ then $\sum a_n(z_1 - z_o)^n$ converges absolutely when $z_1 \in \overline{D}(z_o; r)$, so $\sum |a_n| r^n$ converges, contradicting the definition of R. ■

Corollary:

A power series $\sum_{n=0}^{\infty} a_n(z - z_o)^n$ is an analytic function on the inside of its circle of convergence.

Proof:

Combine power series convergence theorem with analytic convergence theorem. ■

Theorem (Differentiation of Power Series)

Let $f(z) = \sum_{n=0}^{\infty} a_n(z - z_o)^n$ be the analytic function defined inside its circle of convergence $D(z_o; R)$. Then $f'(z) = \sum_{n=1}^{\infty} n a_n(z - z_0)^{n-1}$ and this series has the same circle of convergence as $f(z)$. Furthermore, $a_n = \frac{f^{(n)}(z_o)}{n!}$.

Proof:
By the Analytic Convergence Theorem,

$$f'(z) = \sum_{n=1}^{\infty} na_n(z - z_o)^{n-1}$$

converges on $D(z_o; R)$. It remains to show that it diverges on $|z - z_o| > R$. Let $z_1 \in \mathbb{C}$ be such that $r_0 = |z_1 - z_o| > R$. Then $na_n r_0^{n-1}$ is bounded in absolute value. Hence

$$a_n r_0^n = (na_n r_0^{n-1})\frac{r_0}{n}$$

is also bounded, so $\sum_{n=1} a_n(z - z_o)^n$ converges by the Abel-Weierstrass Lemma for all z such that $R < |z - z_o| < r_0$. This contradicts the definition of R being the radius of convergence of the original power series. Hence the radius of convergence of f' must also be R. To compute a_n, set $z = z_o$ in the formula $f(z) = \sum a_n(z - z_o)^n$ to find $a_n = f(z_o)$. By induction, $f^{(n)}(z_o) = n!a_n + \sum k(k-1)\ldots(k-n+1)(z - z_o)^{k-n}$ and setting $z = z_0$, we get $f^{(n)}(z_o) = n!a_n$. ■

Corollary (uniqueness):
Power series expansions around the same centre are unique, i.e. if $\sum a_n(z - z_o)^n = f(z) = \sum b_n(z - z_o)^n$ for all z in some disk $D(z_o; r)$ with $r > 0$, then $a_n = b_n$ for all $n = 0, 1, 2, , \ldots$.

Calculating the Radius of Convergence
Let $\sum_{n=0}^{\infty} a_n(z - z_o)^n$ be a power series and let R be its radius of convergence.
Ratio Test. If $\lim_{n\to\infty} \frac{|a_n|}{|a_{n+1}|}$ exists, then it is equal to R.
Root Test. If $\rho = \lim_{n\to\infty} \sqrt[n]{|a_n|}$ exists, then $R = 1/\rho$ ($R = \infty$ when $\rho = 0$, $R = 0$ when $\rho = \infty$).

Examples
For $\sum_{n=0}^{\infty} z^n$ has radius of convergence 1, $\sum_{n=0}^{\infty} \frac{z^n}{e}$ has radius of convergence e.

Exercise
Find the radii of convergence for $\sum_{n=0}^{\infty} \frac{2^n}{n^2} z^n$ and $\sum_{n=0}^{\infty} z^{n!}$.

Theorem (Taylor)
Let $f : \Omega \to \mathbb{C}$ be analytic in Ω (Ω open). Let $z_o \in \Omega$ and let $0 \le r \le \infty$ be such that $D(z_o; r) \subseteq \Omega$. Then for every $z \in D(z_o; r)$, the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(z_o)}{n!}(z - z_o)^n$$

converges on $D(z_o; r)$, and we have

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_o)}{n!}(z - z_o)^n$$

Lemma:
If $\sum g_n(z)$ converges uniformly on an open set Ω and $h(z)$ is bounded on Ω, prove that $\sum h(z)g_n(z)$ converges uniformly on Ω to $h(z)\sum g_n(z)$.
Proof:
Exercise.

Proof of Taylor's Theorem:
Let $0 < \sigma < r$. By Cauchy's Integral Formula, for every $z \in D(z_o; \sigma)$,

$$f(z) = \frac{1}{2\pi i}\int_{C(z_o;\sigma)} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Notice that $|z - z_o| < |\zeta - z_o|$ for $\zeta \in C(z_o;\sigma)$, so $\left|\frac{z-z_o}{\zeta-z_o}\right| < 1$. In view of this, we can use the geometric series:

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_o} \cdot \frac{1}{1 - \frac{z-z_o}{\zeta-z_o}} = \frac{1}{\zeta - z_o} \sum_{n=0}^{\infty} \left(\frac{z - z_o)}{\zeta - z_o} \right)^n$$

so that

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{C(z_o;\sigma)} \frac{f(\zeta)}{\zeta - z_o} \sum_{n=0}^{\infty} \left(\frac{z - z_o}{\zeta - z_o} \right)^n d\zeta \\ &= \frac{1}{2\pi i} \int_{C(z_o;\sigma)} \sum_{n=0}^{\infty} \frac{f(\zeta)(z - z_o)^n}{(\zeta - z_o)^{n+1}} d\zeta \end{aligned}$$

We want to interchange this integral and summation, for which we need uniform convergence. For this, we need to convince ourselves that $\frac{f(\zeta)(z-z_o)^n}{(\zeta-z_o)^{n+1}}$ converges to $\frac{f(\zeta)}{\zeta-z}$ on $C(z_o;\sigma)$. Since the curve $C(z_o;\sigma)$ stays away from the boundary of the disk of convergence $D(z_o;r)$, the series $\sum_{n=0}^{\infty}\left(\frac{z-z_o}{\zeta-z_o}\right)^n$ converges uniformly on $C(z_o;\sigma)$ with z fixed. $\frac{f(\zeta)}{\zeta-z_o}$ is a continuous function of ζ on $C(z_o;\sigma)$ (since $z_o \neq \zeta$ always), so it is bounded, since this set is compact. Thus, by the lemma, $\sum \frac{f(\zeta)(z-z_o)^n}{(\zeta-z_o)^{n+1}}$ converges uniformly on $C(z_o;\sigma)$ to $\frac{f(\zeta)}{\zeta-z}$. Hence

$$\begin{aligned} f(z) = \frac{1}{2\pi i} \int_{C(z_o;\sigma)} \sum_{n=0}^{\infty} \frac{f(\zeta)(z - z_o)^n}{(\zeta - z_o)^{n+1}} d\zeta &= \sum_{n=0}^{\infty} \frac{1}{2\pi i} \int_{C(z_o;\sigma)} \frac{f(\zeta)(z - z_o)^n}{(\zeta - z_o)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} \frac{(z - z_o)^n}{2\pi i} \int_{C(z_o;\sigma)} \frac{f(\zeta)}{(\zeta - z_o)^{n+1}} d\zeta \\ &\overbrace{=}^{\text{Cauchy's Diff. Formula}} \sum_{n=0}^{\infty} \frac{f^{(n)}(z_o)}{n!}(z - z_o)^n \end{aligned}$$

Since σ was arbitrary, this representation works for every $D(z_o; sigma) \subseteq D(z_o;r) \subseteq \Omega$, so we have the result since $D(z_o;r)$ is open. ■

Corollary (Analytic and Holomorphic Coincide):
Let Ω be open in $\mathbb{C}$ and let $f : \Omega \to \mathbb{C}$. Then f is holomorphic in Ω if and only if for every $z_o \in \Omega$ there is $r > 0$ such that $D(z_o;r) \subseteq \Omega$ and f is equal to a convergent power series on $D(z_o;r)$.
Proof:
($\Longrightarrow$:) By Taylor's Theorem, if f is holomorphic at z_o, then by Taylor's Theorem, f is equal to its Taylor series on every open disk $D(z_o;r)$ such that $D(z_o;r) \subseteq \Omega$.
($\Longleftarrow$:) Assume that f is equal to a convergent power series on $D(z_o;r) \subseteq \Omega$, with $z_o \in \Omega$. Then, $D(z_o;r) \subseteq D(z_o;R)$ where $D(z_o;R)$ is the disk of convergence of f's power series. Since every power series is holomorphic in its disk of convergence, this shows f is holomorphic on $D(z_o;r)$. ■

Some Examples
Let us derive Taylor series for well-known analytic functions, such as e^z, $\sin(z)$, $\cos(z)$, $\log(1+z)$, $(1+z)^\alpha$ for $\alpha \in \mathbb{C}$ fixed, $1/(1-z)$, etc.

For $f(z) = e^z$, we have $f^{(n)} = e^z$, so $f^{(n)}(0) = 1$, so,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(z - 0)^n = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

We have $\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$, so from above, we have

$$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!}, e^{-iz} = \sum_{n=0}^{\infty} \frac{(-iz)^n}{n!}$$

implying

$$\cos(z) = \sum_{n=0}^{\infty} \frac{i^n + (-i)^n}{2n!} z^n = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}$$

Theorem (Laurent):
Suppose f is analytic around some point z_o, but not necessarily z_o itself (for instance, z^{-1}). Let $r_1 \geq 0, r_2 > r_1$, and $z_o \in \mathbb{C}$. Set $\Omega = \{z \in \mathbb{C} : r_1 < |z_0 - z| < r_2\}$. We allow $r_1 = 0$ or $r_2 = \infty$ or both. Then $f : \Omega \to \mathbb{C}$ is analytic in Ω and has a Laurent series on Ω centered at z_0:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_o)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_o)^n}$$

where both series on the right converge absolutely on Ω and uniformly on every set $B(z_o; \rho_1, \rho_2) = \{z \in \mathbb{C} : \rho_1 \leq |z_o - z| \leq \rho_2\}$ where $r_1 < \rho_1 < \rho_2 < r_2$. Furthermore, for every r such that $r_1 < r < r_2$,

$$a_n = \frac{1}{2\pi i} \int_{C(z_o;r)} \frac{f(\zeta)}{(\zeta - z_o)^{n+1}} d\zeta$$

$$b_n = \frac{1}{2\pi i} \int_{C(z_o;r)} f(\zeta)(\zeta - z_o)^{n-1} d\zeta$$

Any pointwise convergent expansion of this form equals the Laurent series on Ω at z_o, i.e. it is unique.

Example
Uniqueness depends on Ω and z_o. For $|z| > 1$ (centered at 0), with $f(z) = 1/(z(z-1))$, we have

$$f(z) = \frac{1}{z(z-1)} = \frac{1}{z^2(1 - 1/z)} = \frac{1}{z^2} \sum_{n=0}^{\infty} \frac{1}{z^n}$$

using a convergent geometric series.

For $0 < |z| < 1$, again using a convergent geometric series, $f(z) = \frac{1}{z(z-1)} = \frac{-1}{z} \sum_{n=0}^{\infty} z^n$.

Exercise 1 (Lemma for the proof):
Let $z_0 \in \mathbb{C}$ and let ρ_1, ρ_2 be such that $0 < \rho_1 < \rho_2$. Let f be analytic in an open set Ω, which contains $\overline{D}(z_0; \rho_2) \setminus D(z_0; \rho_1)$. Notice that the circles $C(z_0; \rho_1)$ and $C(z_0; \rho_2)$, which we assume to be oriented counterclockwise, are contained inside Ω. Prove that

$$f(z) = \frac{1}{2\pi i} \int_{C(z_o;\rho_2)} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{C(z_o;\rho_1)} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Exercise 2:
Let $z_0, b_0, b_1, \cdots \in \mathbb{C}$ and $R \geq 0$. Suppose that the series $\sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$ converges for every z such that $|z - z_0| > R$. Prove that, for every $r > R$, the series above converges uniformly on set $\{z \in \mathbb{C} : |z - z_0| > r\}$. (Hint: Think how you can adapt the Abel-Weierstrass Lemma and Weierstrass M-test to this special case)

Proof of Laurent's Theorem:
We begin by showing uniform convergence on $B(z_o; \rho_1, \rho_2)$. Since all circles $C(z_o; r)$ with $r_1 < r < r_2$ are homotopic to each other in Ω, we can choose $\tilde{\rho}_1$ and $\tilde{\rho}_2$ so that $r_1 < \tilde{\rho}_1 < \rho_1 < \rho_2 < \tilde{\rho}_2 < r_2$ and put

$$a_n = \frac{1}{2\pi i} \int_{C(z_o;\tilde{\rho}_1)} \frac{f(\zeta)}{(\zeta - z_o)^{n+1}} d\zeta$$

$$b_n = \frac{1}{2\pi i} \int_{C(z_o;\tilde{\rho}_2)} f(\zeta)(\zeta - z_o)^{n-1} d\zeta$$

By exercise 1, for $z \in B(z_o; \rho_1, \rho_2)$ we have

$$f(z) = \frac{1}{2\pi i}\int_{C(z_o;\tilde\rho_2)} \frac{f(\zeta)}{\zeta - z}d\zeta - \frac{1}{2\pi i}\int_{C(z_o;\tilde\rho_1)} \frac{f(\zeta)}{\zeta - z}d\zeta$$

Now, for $B(z_o; \rho_1, \rho_2)$ and for $\zeta \in C(z_o; \tilde\rho_2)$, we have $|\frac{z-z_o}{\zeta-z_o}| < 1$. Hence we can use geometric series expansion:

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_o}\frac{1}{1 - \frac{z-z_o}{\zeta-z_o}} = \frac{1}{\zeta - z_o}\sum_{n=0}^{\infty}\left(\frac{z - z_o}{\zeta - z_o}\right)^n$$

as in the proof of Taylor's theorem, we can show that $\frac{f(\zeta)}{\zeta-z_o}\sum_{n=0}^{\infty}\left(\frac{z-z_o}{\zeta-z_o}\right)^n$ converges uniformly to $\frac{f(\zeta)}{\zeta-z}$ on $C(z_o; \tilde\rho_2)$. Hence we can interchange integration and summation:

$$\frac{1}{2\pi i}\int_{C(z_o;\tilde\rho_2)} \frac{f(\zeta)}{\zeta - z}d\zeta = \sum_{n=0}^{\infty}\frac{1}{2\pi i}\int_{C(z_o;\tilde\rho_2)}\frac{f(\zeta)}{(\zeta - z_o)^{n+1}}d\zeta(z - z_o)^n = \sum_{n=0}^{\infty} a_n(z - z_o)^n$$

Also, for $z \in B(z_o; \rho_1, \rho_2)$ and $\zeta \in C(z_o; \tilde\rho_1)$, $|\frac{\zeta-z_o}{z-z_o}| < 1$, hence we can use geometric series again:

$$-\frac{1}{\zeta - z} = \frac{1}{z - z_o}\frac{1}{1 - \frac{\zeta-z_o}{z-z_o}} = \frac{1}{z - z_o}\sum_{n=0}^{\infty}\left(\frac{\zeta - z_o}{z - z_o}\right)^n$$

As in the proof of Taylor's theorem, we can show that $\frac{f(\zeta)}{z-z_o}\sum_{n=0}^{\infty}\left(\frac{\zeta-z_o}{z-z_o}\right)^n$ converges uniformly to $-\frac{f(\zeta)}{\zeta-z}$ on $C(z_o; \tilde\rho_1)$. Hence we can interchange integration and summation:

$$\frac{-1}{2\pi i}\int_{C(z_o;\tilde\rho_1)} \frac{f(\zeta)}{\zeta - z}d\zeta = \sum_{n=1}^{\infty}\frac{1}{2\pi i}\int_{C(z_o;\tilde\rho_1)} f(\zeta)(\zeta - z_o)^{n-1}d\zeta\frac{1}{(z - z_o)^n} = \sum_{n=1}\frac{b_n}{(z - z_o)^n}$$

(where we shift an index up to start at $n = 1$) giving existence. The series $\sum_{n=0}^{\infty} a_n(z-z_o)^n$ converges on $C(z_o; \tilde\rho_2)$, so it follows by Abel-Weierstrass that it convrges on strictly smaller disks. Also, $\sum_{n=1}^{\infty}\frac{b_n}{(z-z_o)^n}$ converges outside $C(z_o; \tilde\rho_2)$, so it follows from A4Q6b) that it converges on strictly larger disks.

For uniqueness, suppose that $f(z) = \sum_{n=0} a_n(z - z_o)^n + \sum_{n=1}\frac{b_n}{(z-z_o)^n}$. If this converges in Ω, then it must converge uniformly on $C(z_o; r)$ for $r_1 < r < r_2$. But then

$$\frac{f(z)}{(z - z_o)^{k+1}} = \sum_{n=0}^{\infty} a_n(z - z_o)^{n-k-1} + \sum_{n=1}^{\infty}\frac{b_n}{(z - z_o)^{n+k+1}}$$

also converges uniformly on $C(z_o; r)$, hence we can interchange integral and summation signs:

$$\int_{C(z_o;r)}\frac{f(z)}{(z - z_o)^{k+1}}dz = \sum_{n=0}^{\infty} a_n\int_{C(z_o;r)}(z - z_o)^{n-k-1}dz + \sum_{n=1} b_n\int_{C(z_o;r)}\frac{1}{(z - z_o)^{n+k+1}}dz$$

Since $\int_{C(z_o;r)}(z - z_o)^m dz = 0$ if $m \neq -1$ and $2\pi i$ if $m = -1$ (by A4Q6(c)), for $k \geq 0$, we have

$$\int_{C(z_o;r)}\frac{f(z)}{(z - z_o)^{k+1}}dz = 2\pi i a_k$$

In turn, for $k = -n$ with $n \geq 1$, we have

$$\int_{C(z_o;r)}\frac{f(z)}{(z - z_o)^{k+1}}dz = 2\pi i b_{-k}$$

which is the same as

$$\int_{C(z_o;r)} f(z)(z-z_o)^{n-1}dz = 2\pi b_n$$

proving the coefficients are uniquely determined. ■

Definition
Let f be analytic on an open set $\Omega \neq \mathbb{C}$. A point $z_o \in \mathbb{C} \setminus \Omega$ is called an isolated singularity of f if and only if there exists $\varepsilon > 0$ such that f is analytic in $D(z_o;\varepsilon) \setminus \{z_o\}$. An example is z^{-1}.

Note: When z_o is an isolated singularity, a (unique) Laurent series on $D(z_o;\varepsilon) \setminus \{z_o\}$ centered at z_o exists:

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_o)^n + \sum_{n=1} \frac{b_n}{(z-z_o)^n}$$

Definition
We refer to b_1 in the above series as the residue of f at z_o written $b_1 = \text{Res}(f, z_o)$.

Example
The residue of z^{-1} at 0 is 1.

Definition Let z_o be an isolated singularity of f. z_o is a pole if and only if all but finitely many b_n's are zero, otherwise it is called an essential singularity.

Definition
z_o is a removable singularity if and only if all b_n's are zero.

Definition
Let z_o be a pole that is not a removable singularity. Let k be the largest positive integer such that $b_k \neq 0$. Then z_o is a pole of (finite) order k. When $k = 1$, we also say z_o is a simple pole.

Definition
A function f is said to be meromorphic in Ω if and only if it is analytic in Ω except for poles in Ω.

Corollary of Laurent's Theorem (Residue)
Let f be analytic in Ω, $\Omega \neq \mathbb{C}$, and let z_o be an isolated singularity of f. Let $r > 0$ be such that $\overline{D}(z_o;r) \setminus \{z_o\}$ is contained in Ω. Then the residue b_1 of f can be calculated via the formula

$$\int_{C(z_o;r)} f(z)dz = b_1(2\pi i)$$

Proof:
This follows by the formula for b_n in Laurent's Theorem. ■

Theorem (Characterization of Singularities)
Let z_o be an isolated singularity of a function f which is analytic in Ω, $\Omega \neq \mathbb{C}$.

Removable Singularities:
z_o is a removable singularity if and only if any one of the following holds:
(1) f is bounded in a deleted neighborhood of z_o
(2) $\lim_{z\to z_o} f(z)$ exists.
(3) $\lim_{z\to z_o}(z-z_o)f(z) = 0$.

Simple Pole
z_o is a simple pole if and only if $\lim_{z\to z_o}(z-z_o)f(z)$ exists and is not equal to zero. Furthermore, $\lim_{z\to z_o}(z-z_o)f(z) = \text{Res}(f, z_o)$.

Pole of Order $\leq k$

z_o is a pole of order $\leq k$ (or possibly a removable singularity) if and only if any of the following holds:
(1) there exists $M > 0 \in \mathbb{R}$ and $k \in \mathbb{N}$ such that $|f(z)| \leq \frac{M}{|z-z_o|^k}$ for all z in a deleted neighborhood of z_o.
(2) $\lim_{z\to z_o}(z-z_o)^k f(z)$ exists
(3) $\lim_{z\to z_o}(z-z_o)^{k+1} f(z) = 0$.

Pole of Order $k \geq 1$

z_o is a pole of order $k \geq 1$ if and only if there is a function $\phi(z)$ analytic in $D(z_o;r)\setminus\{z_o\}$ for some $r > 0 \in \mathbb{R}$ such that $\phi(z_o) \neq 0$ and

$$f(z) = \frac{\phi(z)}{(z-z_o)^k}$$

for all $z \in D(z_o;r)\setminus\{z_o\}$.
Proof:
For removable singularities, let z_o be a removable singularity of f. Then the Laurent series of $f(z)$ on $D(z_o;\delta)\setminus\{z_o\}$ for some $\delta > 0$ looks like

$$f(z) = \sum TODO$$

Example

Find the Laurent expansion of $f(z) = \frac{z}{z^2+1}$ on $\Omega = \{z \in \mathbb{C} : 0 < |z-z_o| < 2\}$, where $z_o = i$.
Proof:
We have

$$\begin{aligned} f(z) &= \frac{z}{(z+i)(z-i)} = \frac{1}{2}\frac{1}{z+i} + \frac{1}{2}\frac{1}{z-i} \\ &= \frac{1}{2}\frac{1}{2i+(z-i)} + \frac{1}{2}\frac{1}{z-i} \\ &= \frac{1}{4i}\frac{1}{1+\frac{z-i}{2i}} + \frac{1}{2}\frac{1}{z-i} \\ &\overbrace{=}^{\text{geom. series}} \frac{1}{4i}\sum_{n=0}^{\infty}(-1)^n(\frac{z-i}{2i})^n + \frac{1}{2}(z-i)^{-1} \\ &= \frac{(1/2)}{z-i} + \sum_{n=0}^{\infty}\frac{(-1)^n}{2(2i)^{n+1}}(z-i)^n \end{aligned}$$

Definition

Let f be analytic in Ω and let $z_o \in \Omega$. We say that f has a zero of order k at z_o if and only if $f^{(k)}(z_o) \neq 0$ and $f^{(n)}(z_o) = 0$ for all $n < k$. In other words, the Taylor series of f in a neighborhood of z_o is:

$$f(z) = \sum_{n=k}^{\infty}\frac{f^{(n)}(z_o)}{n!}(z-z_o)^n = (z-z_o)^k\sum_{n=k}^{\infty}\frac{f^{(n)}(z_o)}{n!}(z-z_o)^{n-k}$$

Notice that the rightmost series is analytic at z_o and its value at z_o is $\frac{f^{(k)}(z_o)}{k!} \neq 0$.

Exercise

Prove that f has a zero of order k at z_o if and only if $\frac{1}{f}$ has a pole of order k at z_o. Furthermore, prove that if h is analytic at z_o and $h(z_o) \neq 0$, then $\frac{h}{f}$ has a pole of order k at z_o.

Exercise

Let f have a pole of order k at z_o. Prove that $\lim_{z\to z_o}|f(z)| = \infty$.

The above observation is false for essential singularities. Essential singularities are VERY bizarre:
Theorem (Picard):
Let f be analytic in $D(z_o;\delta) \setminus \{z_o\}$ for some $\delta > 0$, and assume that z_o is an essential singularity of f. Then for all $w \in \mathbb{C}$, perhaps one exception, the equation $f(z) = w$ has infinitely many solutions $z \in D(z_o;\delta) \setminus \{z_o\}$.

Residue Calculus
Goal: Learn how to compute $\int_\Gamma f(z)dz$ when f is meromorphic in $\operatorname{Int}\Gamma$, as opposed to analytic.
Application:
Once we know how to do this, we will be able to compute many more real integrals (a la A3Q6).

Theorem (Residue for Jordan Curves):
Let Ω be an open subset of $\mathbb{C}$, let $z_1,\dots,z_n$ be distinct points in Ω. Let f be analytic in $\Omega \setminus \{z_1,\dots,z_n\}$ (so $z_1,\dots,z_n$ are isolated singularities of f). Let Γ be a Jordan curve such that $z_1,\dots,z_n \in \operatorname{Int}\Gamma$ and $\overline{\operatorname{Int}\Gamma} \subseteq \Omega$. Then

$$\int_\Gamma f(z)dz = 2\pi i \sum_{i=1}^{n} \operatorname{Res}(f, z_i)$$

Proof:
Since $\operatorname{Int}\Gamma$ is open, and $z_1,\dots,z_n \in \operatorname{Int}\Gamma$, there exist $r_1,\dots,r_n \in (0,\infty)$ such that $\overline{D}(z_i;r_i) \subseteq \operatorname{Int}\Gamma$ for all i and the disks are pairwise disjoint. Define $\Omega' = \operatorname{Int}\Gamma \setminus \bigcup_i \overline{D}(z_i;r_i)$. This is a $(n+1)$-connected Jordan domain. We orient $\partial\overline{D}(z_i;r_i) = C(z_i;r_i)$ counterclockwise (as in picture in lecture slides). Then from Cauchy's Integral Theorem from k-connected Jordan domains, $\int_{\Gamma \bigcup_i -C(z_i;r_i)} f(z)dz = 0$. TODO FINISH

Theorem (Residue for General Curves):
Let Ω be an open subset of $\mathbb{C}$, let $z_1,\dots,z_n$ be distinct points in Ω. Let f be analytic in $\Omega \setminus \{z_1,\dots,z_n\}$ (so $z_1,\dots,z_n$ are isolated singularities of f). Γ be a piecewise-smooth closed curve in Ω that is homotopic as a closed curve to a point in Ω. Assume that $z_1,\dots,z_n \notin \Gamma$. Then

$$\int_\Gamma f(z)dz = 2\pi i \sum_{i=1}^{n} \operatorname{Res}(f, z_i) I(\Gamma, z_i)$$

where I is the winding number.

Calculating Residues - Removable Singularity
If $g(z)$ and $h(z)$ are analytic and have zeros at z_o of the same order, then $f(z) = \frac{g(z)}{h(z)}$ has a removable singularity at z_0.

Calculating Residues - Simple Poles
Let g and h be analytic at z_0 and assume that $g(z_0) \neq 0$, $h(z_0) = 0$, $h'(z_0) \neq 0$. Then $f(z) = \frac{g(z)}{h(z)}$ has a simple pole at z_0.

Calculating Residues - Double Poles
Let g and h be analytic at z_0 and assume that $g(z_0) \neq 0$, $h(z_0) = 0, h'(z_0) = 0, h''(z_0) \neq 0$. Then $\frac{g(z)}{h(z)}$ has a second order pole at z_0 and $\operatorname{Res}(g/h, z_0) = 2\frac{g'(z_0)}{h''(z_0)} - \frac{2g(z_0)h'''(z_0)}{3h''(z_0)^2}$
More generally if $g(z_0) = 0, g'(z_0) \neq 0$ and $h(z_0) = h'(z_0) = h''(z_0) = 0$ and $h'''(z_0) \neq 0$, then $\frac{g(z)}{h(z)}$ has a second order pole at z_0 and $\operatorname{Res}(g/h, z_0) = 2\frac{g''(z_0)}{h'''(z_0)} - \frac{2}{3}\frac{g(z_0)h''''(z_0)}{h'''(z_0)^2}$

Proposition
Let f have an isolated singularity at z_o and let k be the smallest non-negative integer such that $\lim_{z\to z_o}(z - z_o)^k f(z)$ exists. Then f has a pole of order k at z_o and $\phi(z) = (z - z_o)^k f(z)$ can be

uniquely defined at z_o such that ϕ is analytic at z_o and $\text{Res}(f, z_o) = \frac{\phi^{(k-1)}(z_o)}{(k-1)!}$
Proof:
Since $\lim_{z\to z_o}^k f(z)$, $\phi(z) = (z - z_o)^k f(z)$ has a removable singularity as z_o. Thus in a neighbourhood of z_o, $\phi(z) = (z-z_o)^k f(z) = b_k + b_{k-1}(z-z_o) + \cdots + b_1(z-z_o)^{k-1} + a_o(z-z_o)^k + \ldots$, which is a power series defined at z_o which we can extend ϕ via, and $f(z) = b_k/(z-z_o)^k + \cdots + b_1/(z-z_o) + a_o + a_1(z - z_O) + \ldots$. If $b_k = 0$, then $\lim_{z\to z_o}(z - z_o)^{k-1} f(z)$ exists, contradicting minimality of k, so we must have $b_k \neq 0$. Thus f has a pole of order k at z_o, and $\phi^{(k-1)}(z) = (k-1)!b_1 + \frac{k!}{1!}a_0(z - z_o) + \frac{(k+1)!}{2!}a_1(z - z_o)^2 + \ldots$, so $\phi^{(k-1)}(z_o) = (k-1)!b_1$, so $\text{Res}(f, z_o) = b_1 = \frac{\phi^{(k-1)}(z_o)}{(k-1)!}$. ■

Using residues to compute real integrals

Example

Consider the integral $\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}dx$. For $1 \le x < \infty$, we have $0 < \frac{x^2}{x^4+1} \le \frac{x^2}{x^4} = \frac{1}{x^2}$, the rightmost of whose improper integral converges, so by comparison $\int_1^{\infty} \frac{x^2}{x^4+1}dx$ converges. As $\frac{x^2}{x^4+1}$ is even, a change of variables shows $\int_{-\infty}^{-1} \frac{x^2}{x^4+1}dx$ also converges, as does $\int_{-1}^{1} \frac{x^2}{x^4+1}dx$, so $\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}dx$ converges.

Therefore, since it converges,

$$\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}dx = \lim_{R\to\infty} \int_{-R}^{R} \frac{x^2}{x^4+1}dx$$

Now, let $f(z) = \frac{z^2}{z^4+1}$, which restricts to $\frac{x^2}{x^2+1}$ on the real axis. f has isolated singularities at the solutions of $z^4 + 1 = 0$, namely $z_1 = e^{\pi i/4}$, $z_2 = e^{3\pi i/4}$, $z_3 = e^{5\pi i/4}$, $z_4 = e^{7\pi i/4}$.

Define Γ_R to be the closed contour $\gamma_R + \mu_R$, where γ_R is the line from $-R$ to R, and μ_R is the counterclockwise semicircle from R to $-R$.

We have

$$\int_{\Gamma_R} \frac{z^2}{z^4+1}dz = 2\pi i(\text{Res}(f, z_1) + \text{Res}(f, z_2))$$

We have $f(z) = g(z)/h(z)$, with $g(z) = z^2$, $h(z) = z^4 + 1$, and $h'(z) = 4z^3$. We have $g(z_i) \neq 0$, $h(z_i) = 0$, and $h'(z_i) \neq 0$, so by a theorem above, $\text{Res}(f, z_i) = g(z_i)/h'(z_i) = z_i^2/4z_i^3 = 1/4z_i$. Thus,

$$2\pi i(\text{Res}(f, z_1) + \text{Res}(f, z_2)) = \frac{\pi i}{2}(e^{-\pi i/4} + e^{-3\pi i/4}) = \frac{\pi i}{2}(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}) = \frac{\pi}{\sqrt{2}}$$

Now let us show as $R \to \infty$, the integral over μ_R goes to 0, which we do by an ML-estimate: We have (on the semicircle) $|z^4 + 1| \ge |z|^4 - 1 = R^4 - 1 > \frac{1}{2}R^4$ if $R > 2$, so

$$\left|\frac{z^2}{z^4+1}\right| = \frac{|z|^2}{|z^4+1|} \le \frac{R^2}{1/2R^4} = \frac{2}{R^2}$$

But then by the ML inequality,

$$\left|\int_{\mu_R} \frac{z^2}{z^4+1}dz\right| \le \frac{2\pi R}{R^2} = \frac{2\pi}{R} \overset{R\to\infty}{\to} 0$$

So

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}dx = \int_{\gamma_R} \frac{z^2}{z^4+1}dz &= \int_{\Gamma_R} \frac{z^2}{z^4+1}dz - \int_{\mu_R} \frac{z^2}{z^4+1}dz \\ &= \lim_{R\to\infty} \int_{\Gamma_R} \frac{z^2}{z^4+1}dz - \lim_{R\to\infty} \int_{\mu_R} \frac{z^2}{z^4+1}dz \\ &= \frac{\pi}{\sqrt{2}} - 0 = \frac{\pi}{\sqrt{2}}\end{aligned}$$

Exercise:
$\int_{-\infty}^{\infty} \frac{\cos(ax)}{1+x^2} dx$, $a \neq 0$.

Example:
Use residues to show $\int_1^{\infty} \frac{1}{x\sqrt{x^2-1}} dx = \frac{\pi}{2}$. We'd like to consider $\frac{1}{z\sqrt{z^2-1}}$ but we have to take about the square root term (hi branch cuts). Let's consider $\sqrt{z-1}\sqrt{z+1}$. On $C \setminus (-\infty, 1]$, take $\sqrt{z-1} = \sqrt{|z-1|}e^{i\arg(z-1)/2}$ with $\arg(z-1) \in (-\pi, \pi)$. On $\mathbb{C} \setminus [-1, \infty)$, take $\sqrt{z+1} = \sqrt{|z+1|}e^{i\arg(z+1)/2}$, where $\arg(z+1) \in (0, 2\pi)$. For these choices, $f(z) = \frac{1}{z\sqrt{z^2-1}}$ is analytic on $\mathbb{C} \setminus \mathbb{R}$. Note, if we cross the branch cut, the square root changes sign (i.e. we multiply by -1). If we cross in $[-1, 1]$, both $\sqrt{z-1}$ and $\sqrt{z+1}$ change by a factor of -1, so $\sqrt{z^2-1}$ does not change. So $f(z) = \frac{1}{z\sqrt{z^2-1}}$ is holomorphic on $\mathbb{C} \setminus (-\infty, -1) \cup (1, \infty)$. We draw a keyhole contour with big radius R, small radius ε, and δ the distance between the top and bottom lines. First, show the integral over the partial circle of radius R goes to zero as $R \to \infty$. Show the integral over the partial circles of radius ε go to zero as $\varepsilon \to 0$. For fixed ε and R, the integral over the straight line segments approaches $\int_{1+\varepsilon}^{R} \frac{1}{x\sqrt{x^2-1}} dx$ as $\delta \to 0$.

Exercise
$\int_{-\infty}^{\infty} \frac{1}{1+x^{2n}} dx$ for n a positive integer.

Summation Theorem
Let f be analytic in $\mathbb{C}$ except for finitely many isolated singularities. Let C_N be a square with vertices at $(N + \frac{1}{2}) \times (\pm 1 \pm i)$, $N \in \mathbb{N}$. Suppose that $\lim_{N\to\infty} \int_{C_N} (\pi\cot(\pi z)) f(z) dz = 0$ ($\cot(\pi z)$ has simple poles at the integers). If none of the singularities of f are at integers, then $\lim_{N\to\infty} \sum_{-N}^{N} f(n)$ exists, is finite, and $\lim_{N\to\infty} \sum_{n=-N}^{N} f(n) = -\sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at singularities of } f\}$.
Proof:
By the residue theorem,

$$\int_{C_N} (\pi\cot(\pi z)) f(z) dz = 2\pi i \sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at } -N, \ldots, N\}$$
$$+2\pi i \sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at singularities of } f\}$$

for N sufficiently large so that $\mathrm{Int}(C_N)$ contains all the isolated singularities of f. Now for $g(z) = \cot(\pi z)$ and $h(z) = \sin(\pi z)$, then for every $n \in \mathbb{Z}$, $g(n) \neq 0$, $h(n) = 0$, $h'(n) \neq 0$. Hence $\cot(\pi z) = \frac{g(z)}{h(z)}$ has a simple pole at every integer and $\mathrm{Res}(g, n) = \frac{g(n)}{h'(n)} = \frac{\cos(\pi n)}{\pi\cos(\pi n)} = \frac{1}{\pi}$. Hence $\mathrm{Res}(\pi\cot(\pi z)f(z), n) = \pi f(n)\,\mathrm{Res}(\cot(\pi z), n) = f(n)$. Hence $\sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at } -N, \ldots, N\} = \sum_{n=-N}^{N} f(n)$. Letting $N \to \infty$, we get

$$\underbrace{\lim_{N\to\infty} \int_{C_N} (\pi\cot(\pi z)) f(z) dz}_{\to 0 \text{ by hyp.}} = \lim_{N\to\infty} 2\pi i \sum_{n=-N}^{N} f(n)$$
$$+ \lim_{N\to\infty} \underbrace{2\pi i \sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at singularities of } f\}}_{\text{constant wrt } N}$$

Hence

$$\lim_{N\to\infty} \sum_{n=-N}^{N} f(n) = -\sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at singularities of } f\} \quad \blacksquare$$

Remark:
The proof above shows more generally that when $f(z)$ does have integer singularities,

$$\lim_{N\to\infty} \sum_{n=-N, n \text{ not sing. of } f}^{N} f(n) = -\sum\{\text{residues of } \pi\cot(\pi z)f(z) \text{ at singularities of } f\}$$

Example
Let us evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Take $f(z) = \frac{1}{z^2}$. Let's assume that $\lim_{N\to\infty} \int_{C_N} \pi\cot(\pi z) f(z)dz = 0$ to avoid proving more annoying things. By the Summation Theorem and the remark after,

$$\lim_{N\to\infty} \sum_{n=-N, n\neq 0}^{N} f(n) = -\operatorname{Res}(\pi\cot(\pi z)/z^2, 0)$$

On the other hand, by evenness, $\sum_{n=-N,n\neq 0}^{N} f(n) = \sum_{n=-N,n\neq 0}^{N} = 2\sum_{n=1} \frac{1}{n^2}$.

Hence $\lim_{N\to\infty} 2\sum_{n=1}^{N} \frac{1}{n^2} = -\operatorname{Res}(\frac{\pi\cot(\pi z)}{z^2}, 0)$. Let us find the first few terms in the Laurent expansion of $\cot(z)$ at 0:

$$\cot(z) = \frac{b_1}{z} + a_0 + a_1 z + \dots$$

But $\cos(z) = \sin(z)\cot(z)$, so

$$(1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \dots) = (z - \frac{z^3}{3!} + \frac{z^5}{5!} + \dots)(b_1 z^{-1} + a_0 + a_1 z + \dots)$$

$$= b_1 + a_0 z + (a_1 - \frac{b_1}{6})z^2$$

hence $b_1 = 1, a_0 = 0$, $a_1 - \frac{b_1}{6} = -\frac{1}{2}$, so $a_1 = \frac{-1}{3}$. Thus

$$\frac{\pi\cot(\pi z)}{z^2} = \frac{\pi}{z^2}(\frac{1}{\pi z} - \frac{\pi z}{3} + \dots) = \frac{1}{z^3} - \frac{\pi^2}{3}\frac{1}{z} + \dots$$

Hence $\operatorname{Res}(\frac{\pi\cot(\pi z)}{z^2}, 0) = -\frac{\pi^2}{3}$ and the result follows. ■

Riemann Sphere
Idea: Add the point at ∞ to $\mathbb{C}$, so to obtain an extended plane $\hat{C} = \mathbb{C} \cup \{\infty\}$. We can then say that numbers that are large in absolute value are close to ∞, just like numbers that are small in absolute value are close to zero. We will also be able to say what it means for a function to be holomorphic at ∞. If z_o is a pole of f, we will be able to say that $f(z_o) = \infty$. In summary, in many ways, ∞ on $\hat{C}$ behaves like an ordinary point!

Zeros/Poles/Residues
If there exists $R > 0$ such that f is analytic at z for all $|z| \geq R$, then we can think of f as being analytic at ∞ (in the sense of the Riemann sphere). Define $F(z) = f(1/z)$. We say that f has a pole of order k at infinity if and only if F has a pole of order k at 0, f has a zero of order k at infinity if F has a pole of order k at infinity, and the residue of f at infinity is $\operatorname{Res}(f, \infty) = \operatorname{Res}(\frac{1}{z^2}f(z), 0)$.

Theorem
Suppose there is $R_0 > 0$ such that f is analytic for $|z| > R_0$. If $R > R_0$, then $\int_{C(0;R)} f(z)dz = -2\pi i \operatorname{Res}(f, 0)$.
Proof:
Let $r = 1/R$. If $z \in D(0;r)$, then $1/z \in \mathbb{C} \setminus D(0;r)$. Hence $g(z) = \frac{1}{z^2}f(1/z)$ is analytic in $D(0;r)$, except possibly at 0. By the Residue Theorem,

$$2\pi i \operatorname{Res}(g, 0) = \int_{C(0;r)} \frac{1}{z^2} f(\frac{1}{z})dz = \int_0^{2\pi} \frac{1}{(re^{\theta i})^2} f(\frac{1}{re^{\theta i}})d(re^{\theta i})$$

$$= \int_0^{2\pi} f(Re^{-\theta i})Re^{-\theta i}d\theta = -\int_{C(0;R)} f(z)dz \quad ■$$

Corollary:
Let Γ be a Jordan curve, oriented counter-clockwise. Let f be analytic along Γ and outside Γ, except for only finitely many singularities outside Γ. Then

$$\int_\Gamma f(z)dz = -2\pi i \sum [\text{residues of } f \text{ outside } \gamma, \text{ including } \infty]$$

Lemma (Argument Principle):
Let Γ be a positively oriented Jordan curve. Suppose that f is analytic inside and on Γ, except for a finite number of poles inside Γ, but with no poles or zeros on the curve Γ itself. Then

$$\int_\Gamma \frac{f'(z)}{f(z)}dz = 2\pi i(Z - P)$$

where Z denotes the number of zeroes of f in $\operatorname{Int}\Gamma$ (counting order) and P denotes the number of poles of f in $\operatorname{Int}\Gamma$ (counting order).
Proof:
Assignment 5. ■

Theorem (Rouche for Jordan Curves):
Let Γ be a Jordan curve, oriented counter-clockwise. Let f and g be polynomials such that $|f(z)-g(z)| < |f(z)|+|g(z)|$ for all $z \in \Gamma$. Then $\operatorname{Int}\Gamma$ contains the same number of roots of f and g, counting multiplicity.
Proof:
The above lemma becomes $\int_\Gamma \frac{f'(z)}{f(z)}dz = 2\pi i Z$ for polynomials. Let z_f denote the number of roots of f. Let z_g denote the number of roots of g, both up to multiplicity. By the argument principle, we have

$$\int_\Gamma \frac{f'(z)}{f(z)}dz = 2\pi i z_f$$

and

$$\int_\Gamma \frac{g'(z)}{g(z)} = 2\pi i z_g$$

(notice that the inequality $|f(z) - g(z)| < |f(z)| + |g(z)|$ for all $z \in \Gamma$ ensures that no zeros of f and g lie on Γ itself). We will show that $\int_\Gamma \frac{f'(z)}{f(z)}dz = \int_\Gamma \frac{g'(z)}{g(z)}dz$. Let $\alpha : [0,1] \to \Gamma$ be a piecewise-smooth parameterization of Γ and notice that $f \circ \alpha$ is a parameterization of $f(\Gamma)$. But then

$$\frac{1}{2\pi i}\int_\Gamma \frac{f'(z)}{f(z)}dz = \frac{1}{2\pi i}\int_{f\circ\alpha} \frac{dz}{z} = I(f(\Gamma), 0)$$

Similarly

$$\frac{1}{2\pi i}\int_\Gamma \frac{g'(z)}{g(z)}dz = I(g(\Gamma), 0)$$

We will show that the curves $f(\Gamma)$ and $g(\Gamma)$ are homotopic as closed curves in $\mathbb{C} \setminus \{0\}$ (they are closed since $\alpha(0) = \alpha(1) \implies f(\alpha(0)) = f(\alpha(1))$). Define: $H : [0,1] \times [0,1] \to \mathbb{C} \setminus \{0\}$ by $H(s,t) = (1-t)f(\alpha(s)) + tg(\alpha(s))$. Define $\beta_t : [0,1] \to \mathbb{C}\setminus\{0\}$ by $\beta_t(s) = H(s,t)$ and let Γ_t denote the corresponding curve of this parameterization. Then $\Gamma_0 = f(\Gamma)$ and $\Gamma_1 = g(\Gamma)$. Moreover, notice that $H(s,t) \neq 0$ for all $s, t \in [0,1]$, because of the condition $|f(z)-g(z)| < |f(z)|+|g(z)|$ for all $z \in \Gamma$ since $|f(\alpha(s)) - g(\alpha(s))| < |f(\alpha(s))| + |g(\alpha(s))|$. Thus H is a homotopy.

Define $J(t) = \frac{1}{2\pi i}\int_{\Gamma_t} \frac{dz}{z} = I(\Gamma_t, 0)$, which is an integer for all t (with $J(0)$ and $J(1)$ being what we care about). But J is continuous for all $t \in [0,1]$ because of continuity of our construction of H! Therefore it cannot jump between integers, so $J(0) = J(1)$, so $I(f(\Gamma), 0) = I(g(\Gamma, 0))$, giving the result. ■

Example
How many zeros does $g(z) = z^6 - 4z^5 + z^2 - 1$ have in $D(0;1)$?

Solution:
Let $f(z) = -4z^5$. Then for every $z \in C(0;1)$, $|f(z) - g(z)| = |z^6 + z^2 - 1| \leq |z|^6 + |z|^2 + 1 = 3 < 4 = |-4z^5| \leq |f(z)| + |g(z)|$. Thus by Rouche's Theorem, since $f(z)$ has 5 roots up to multiplicity inside the unit circle, $g(z)$ has 5 roots up to multiplicity inside the unit circle.

Definition
A map $f : \Omega \to \mathbb{C}$ is called conformal at z_0 there exists $\alpha[0, 2\pi)$ and $r > 0$ such that for every curve $\gamma : [-1, 1] \to \Omega$ which is differentiable at 0, with $\gamma(0) = z_0$ and $\gamma'(0) \neq 0$, the curve $\sigma = f \circ \gamma$ is differentiable at 0, $|\sigma'(0)| = r|\gamma'(0)|$ and $\arg \sigma'(0) = \arg \gamma'(0) + \alpha \mod 2\pi$. A map is called conformal if it is conformal at every point in its domain.

The intuition is that a conformal mapping merely rotates and stretches tangent vectors of curves. In particular, it preserves angles between curves.

Theorem (Conformal Mapping Theorem):
If $f : \Omega \to \mathbb{C}$ is analytic at z_0 and if $f'(z_0) \neq 0$, then f is conformal at z_0 with $r = |f'(z_0)|$ and $\alpha = \arg f'(z_0)$.
Proof:
By the chain rule, $\sigma'(0) = f'(z_0) = \gamma'(0)$. Since the left and right hand sides are nonzero, we can write them in polar form:

$$|\sigma'(0)|e^{\arg \sigma'(0)i} = re^{\alpha i}|\gamma'(0)|e^{\arg \gamma'(0)i} = (r|\gamma'(0)|)e^{(\alpha + \arg \gamma'(0))i}$$

Equating absolute values and the arguments mod 2π, $|\sigma'(0)| = r|\gamma'(0)|$ and $\arg \sigma'(0) = \arg \gamma'(0) + \alpha \mod 2\pi$. TODO

TODO missing lecture 34/35
Definition (Riemann Zeta Function)
We define the Riemann zeta function by

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

This is analytic for all $s \in \mathbb{C}$ such that $\text{Re}(s) > 1$.
Proof:
Let $s = \sigma + it$, $\sigma, t \in \mathbb{R}, \sigma > 1$. We show absolute and uniform convergence of our series. Then

$$\sum_{n=1}^{\infty} \frac{1}{n^s} \leq \sum_{n=1}^{\infty} \frac{1}{|n^s|} = \sum_{n=1}^{\infty} \frac{1}{|e^{(\sigma+it)\log n}|}$$

$$= \sum_{n=1}^{\infty} \frac{1}{|e^{\sigma \log n}||e^{it \log n}|} = \sum_{n=1}^{\infty} \frac{1}{n^\sigma}$$

since $\sigma > 1$, the series converges by the p-test, so $\sum_{n=1}^{\infty} n^{-s}$ converges absolutely. It also converges uniformly on $\text{Re}(s) \geq 1 + \delta$ (use Cauchy's criterion or M-test). Therefore, by the analytic convergence theorem, $\zeta(s)$ is analytic on $\text{Re}(s) > 1$. $\blacksquare$.

Before we talk about the analytic continuation of the Riemann zeta function, we need to introduce Theta Functions:

Definition
For $t \in \mathbb{R}$, define the theta function by $\nu(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ (this is a consequence of the Poisson summation formula and Fourier Transforms).

Exercise
Use the functional equation $\nu(t) = t^{-1/2}\nu(1/t)$ to prove that there is $C > 0 \in \mathbb{R}$ such that $\nu(t) \leq Ct^{-1/2}$

as $t \to 0^+$ and that $|\nu(t) - 1| \leq Ce^{-\pi t}$ for all $t \geq 1$.

After defining

$$\hat{\zeta}(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

we'll show

$$\hat{\zeta}(s) = \hat{\zeta}(1-s)$$

for all $s \in \mathbb{C} \setminus \{0, 1\}$.